

# NUCLEAR PHYSICS

## 23.1 Mass defect and nuclear binding energy

Candidates should be able to:

- 1 understand the equivalence between energy and mass as represented by  $E = mc^2$  and recall and use this equation
- 2 represent simple nuclear reactions by nuclear equations of the form  ${}^{14}_7\text{N} + {}^4_2\text{He} \rightarrow {}^{17}_8\text{O} + {}^1_1\text{H}$
- 3 define and use the terms mass defect and binding energy
- 4 sketch the variation of binding energy per nucleon with nucleon number
- 5 explain what is meant by nuclear fusion and nuclear fission
- 6 explain the relevance of binding energy per nucleon to nuclear reactions, including nuclear fusion and nuclear fission
- 7 calculate the energy released in nuclear reactions using  $E = c^2\Delta m$

## 23.2 Radioactive decay

Candidates should be able to:

- 1 understand that fluctuations in count rate provide evidence for the random nature of radioactive decay
- 2 understand that radioactive decay is both spontaneous and random
- 3 define activity and decay constant, and recall and use  $A = \lambda N$
- 4 define half-life
- 5 use  $\lambda = 0.693 / t_{1/2}$
- 6 understand the exponential nature of radioactive decay, and sketch and use the relationship  $x = x_0 e^{-\lambda t}$ , where  $x$  could represent activity, number of undecayed nuclei or received count rate

**1. M/J/07/Q6**

- (a)** Define the *decay constant* of a radioactive isotope.

.....  
.....  
..... [2]

- (b)** Strontium-90 is a radioactive isotope having a half-life of 28.0 years. Strontium-90 has a density of  $2.54 \text{ g cm}^{-3}$ .

A sample of Strontium-90 has an activity of  $6.4 \times 10^9 \text{ Bq}$ . Calculate

- (i)** the decay constant  $\lambda$ , in  $\text{s}^{-1}$ , of Strontium-90,

$\lambda = \dots\dots\dots \text{s}^{-1}$  [2]

- (ii)** the mass of Strontium-90 in the sample,

mass =  $\dots\dots\dots \text{g}$  [4]

(iii) the volume of the sample.

volume = ..... cm<sup>3</sup> [1]

(c) By reference to your answer in (b)(iii), suggest why dust that has been contaminated with Strontium-90 presents a serious health hazard.

.....

.....

..... [2]

2. O/N/07/Q7

(a) Explain what is meant by the *binding energy* of a nucleus.

.....  
 .....[1]

(b) Fig. 7.1 shows the variation with nucleon number (mass number)  $A$  of the binding energy per nucleon  $E_B$  of nuclei.

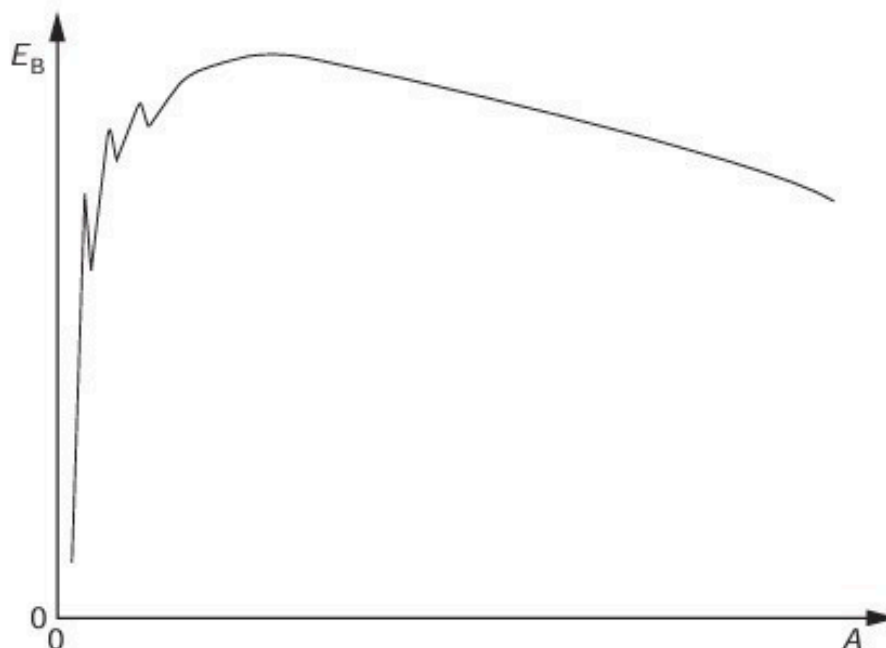
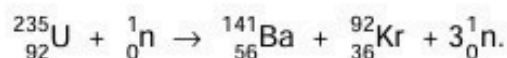


Fig. 7.1

One particular fission reaction may be represented by the nuclear equation



(i) On Fig. 7.1, label the approximate positions of

1. the uranium ( ${}_{92}^{235}\text{U}$ ) nucleus with the symbol U,
2. the barium ( ${}_{56}^{141}\text{Ba}$ ) nucleus with the symbol Ba,
3. the krypton ( ${}_{36}^{92}\text{Kr}$ ) nucleus with the symbol Kr.

[2]

(ii) The neutron that is absorbed by the uranium nucleus has very little kinetic energy. Explain why this fission reaction is energetically possible.

.....  
 .....  
 .....[2]



- (c) Barium-141 has a half-life of 18 minutes. The half-life of Krypton-92 is 3.0 s.  
In the fission reaction of a mass of Uranium-235, equal numbers of barium and krypton nuclei are produced.  
Estimate the time taken after the fission of the sample of uranium for the ratio

$$\frac{\text{number of Barium-141 nuclei}}{\text{number of Krypton-92 nuclei}}$$

to be approximately equal to 8.

time = ..... s [3]

**3. M/J/09/Q9**

- (a) A sample of a radioactive isotope contains  $N$  nuclei at time  $t$ . At time  $(t + \Delta t)$ , it contains  $(N - \Delta N)$  nuclei of the isotope.

For the period  $\Delta t$ , state, in terms of  $N$ ,  $\Delta N$  and  $\Delta t$ ,

- (i) the mean activity of the sample,

activity = ..... [1]

- (ii) the probability of decay of a nucleus.

probability = ..... [1]

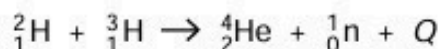
- (b) A cobalt-60 source having a half-life of 5.27 years is calibrated and found to have an activity of  $3.50 \times 10^5$  Bq. The uncertainty in the calibration is  $\pm 2\%$ .

Calculate the length of time, in days, after the calibration has been made, for the stated activity of  $3.50 \times 10^5$  Bq to have a maximum possible error of 10%.

time = ..... days [4]

**4. O/N/09/41/Q8**

The controlled reaction between deuterium ( ${}^2_1\text{H}$ ) and tritium ( ${}^3_1\text{H}$ ) has involved ongoing research for many years. The reaction may be summarised as



where  $Q = 17.7\text{ MeV}$ .

Binding energies per nucleon are shown in Fig. 8.1.

|                   | binding energy per nucleon<br>/MeV |
|-------------------|------------------------------------|
| ${}^2_1\text{H}$  | 1.12                               |
| ${}^1_0\text{n}$  | –                                  |
| ${}^4_2\text{He}$ | 7.07                               |

**Fig. 8.1**

- (a)** Suggest why binding energy per nucleon for the neutron is not quoted.

.....  
 ..... [1]

- (b)** Calculate the mass defect, in kg, of a helium  ${}^4_2\text{He}$  nucleus.

mass defect = ..... kg [3]

- (c) (i)** State the name of the type of reaction illustrated by this nuclear equation.

..... [1]

- (ii)** Determine the binding energy per nucleon, in MeV, of tritium ( ${}^3_1\text{H}$ ).

5. O/N/09/42/Q8

- (a) State what is meant by the *decay constant* of a radioactive isotope.

.....

.....

..... [2]

- (b) Show that the decay constant  $\lambda$  is related to the half-life  $t_{\frac{1}{2}}$  by the expression

$$\lambda t_{\frac{1}{2}} = 0.693.$$

[3]

- (c) Cobalt-60 is a radioactive isotope with a half-life of 5.26 years ( $1.66 \times 10^8$  s).

A cobalt-60 source for use in a school laboratory has an activity of  $1.8 \times 10^5$  Bq.

Calculate the mass of cobalt-60 in the source.

mass = ..... g [3]

**6. M/J/10/42/Q8**

Americium-241 is an artificially produced radioactive element that emits  $\alpha$ -particles.  
A sample of americium-241 of mass  $5.1\ \mu\text{g}$  is found to have an activity of  $5.9 \times 10^5\ \text{Bq}$ .

**(a)** Determine, for this sample of americium-241,

**(i)** the number of nuclei,

number = ..... [2]

**(ii)** the decay constant,

decay constant = .....  $\text{s}^{-1}$  [2]

**(iii)** the half-life, in years.

half-life = ..... years [2]

**(b)** Another radioactive element has a half-life of approximately 4 hours.  
Suggest why measurement of the mass and activity of a sample of this element is not appropriate for the determination of its half-life.

.....  
..... [1]

7. M/J/11/41/Q8

- (a) The variation with nucleon number  $A$  of the binding energy per nucleon  $B_E$  of nuclei is shown in Fig. 8.1.

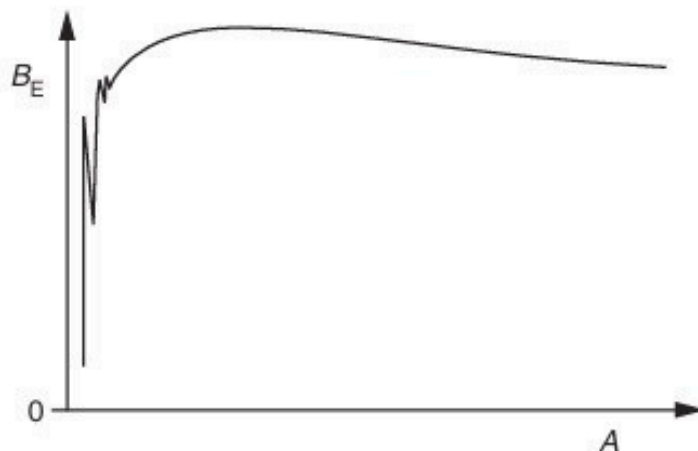


Fig. 8.1

On Fig. 8.1, mark the approximate positions of

- (i) iron-56 (label this point Fe), [1]
- (ii) zirconium-97 (label this point Zr), [1]
- (iii) hydrogen-2 (label this point H). [1]

- (b) (i) State what is meant by *nuclear fission*.

.....  
 .....  
 ..... [2]

- (ii) By reference to Fig. 8.1, explain how fission is energetically possible.

.....  
 .....  
 .....  
 ..... [2]

**8. M/J/11/42/Q8**

**(a)** State what is meant by the *binding energy* of a nucleus.

.....

.....

..... [2]

**(b)** Show that the energy equivalence of 1.0 u is 930 MeV.

[3]

**(c)** Data for the masses of some particles and nuclei are given in Fig. 8.1.

|                                       | mass / u |
|---------------------------------------|----------|
| proton                                | 1.0073   |
| neutron                               | 1.0087   |
| deuterium ( ${}^2_1\text{H}$ )        | 2.0141   |
| zirconium ( ${}^{97}_{40}\text{Zr}$ ) | 97.0980  |

**Fig. 8.1**

Use data from Fig. 8.1 and information from **(b)** to determine, in MeV,

**(i)** the binding energy of deuterium,

binding energy = ..... MeV [2]

(ii) the binding energy **per nucleon** of zirconium.

binding energy per nucleon = ..... MeV [3]



**9. O/N/11/41/Q8**

The isotope phosphorus-33 ( $^{33}_{15}\text{P}$ ) undergoes  $\beta$ -decay to form sulfur-33 ( $^{33}_{16}\text{S}$ ), which is stable.

The half-life of phosphorus-33 is 24.8 days.

**(a) (i)** Define radioactive *half-life*.

.....  
.....  
..... [2]

**(ii)** Show that the decay constant of phosphorus-33 is  $3.23 \times 10^{-7} \text{ s}^{-1}$ .

[1]

**(b)** A pure sample of phosphorus-33 has an initial activity of  $3.7 \times 10^6 \text{ Bq}$ .

Calculate

**(i)** the initial number of phosphorus-33 nuclei in the sample,

number = ..... [2]

**(ii)** the number of phosphorus-33 nuclei remaining in the sample after 30 days.

number = ..... [2]

- (c) After 30 days, the sample in (b) will contain phosphorus-33 and sulfur-33 nuclei.  
Use your answers in (b) to calculate the ratio

$$\frac{\text{number of phosphorus-33 nuclei after 30 days}}{\text{number of sulfur-33 nuclei after 30 days}} .$$

ratio = .....[2]

**10. O/N/11/43/Q8**

Radon-222 is a radioactive element having a half-life of 3.82 days.

Radon-222, when found in atmospheric air, can present a health hazard. Safety measures should be taken when the activity of radon-222 exceeds 200 Bq per cubic metre of air.

**(a) (i)** Define radioactive *decay constant*.

.....  
.....  
..... [2]

**(ii)** Show that the decay constant of radon-222 is  $2.1 \times 10^{-6} \text{ s}^{-1}$ .

[1]

**(b)** A volume of  $1.0 \text{ m}^3$  of atmospheric air contains  $2.5 \times 10^{25}$  molecules.

Calculate the ratio

$$\frac{\text{number of air molecules in } 1.0 \text{ m}^3 \text{ of atmospheric air}}{\text{number of radon-222 atoms in } 1.0 \text{ m}^3 \text{ of atmospheric air}}$$

for the minimum activity of radon-222 at which safety measures should be taken.

ratio = ..... [3]

11. M/J/12/41/Q9

- (a) (i) State what is meant by the *decay constant* of a radioactive isotope.

.....

.....

..... [2]

- (ii) Show that the decay constant  $\lambda$  and the half-life  $t_{\frac{1}{2}}$  of an isotope are related by the expression

$$\lambda t_{\frac{1}{2}} = 0.693.$$

[3]

- (b) In order to determine the half-life of a sample of a radioactive isotope, a student measures the count rate near to the sample, as illustrated in Fig. 9.1.

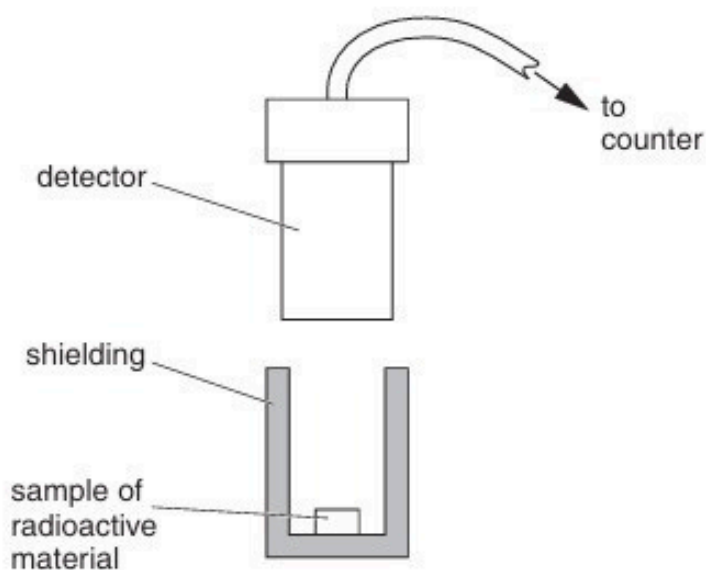


Fig. 9.1

Initially, the measured count rate is 538 per minute. After a time of 8.0 hours, the measured count rate is 228 per minute.

Use these data to estimate the half-life of the isotope.

half-life = ..... hours [3]

- (c) The accepted value of the half-life of the isotope in (b) is 5.8 hours.  
The difference between this value for the half-life and that calculated in (b) cannot be explained by reference to faulty equipment.

Suggest two possible reasons for this difference.

1. ....  
.....
2. ....  
.....

[2]

**12. M/J/12/42/Q8**

The element strontium has at least 16 isotopes. One of these isotopes is strontium-89. This isotope has a half-life of 52 days.

- (a)** State what is meant by *isotopes*.

.....  
.....  
..... [2]

- (b)** Calculate the probability per second of decay of a nucleus of strontium-89.

probability = .....  $\text{s}^{-1}$  [3]

- (c)** A laboratory prepares a strontium-89 source.

The activity of this source is measured 21 days after preparation of the source and is found to be  $7.4 \times 10^6 \text{ Bq}$ .

Determine, for the strontium-89 source at the time that it was prepared,

- (i)** the activity,

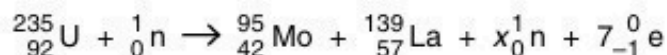
activity = ..... Bq [2]

- (ii)** the mass of strontium-89.

mass = ..... g [2]

13. O/N/12/43/Q8

When a neutron is captured by a uranium-235 nucleus, the outcome may be represented by the nuclear equation shown below.



- (a) (i) Use the equation to determine the value of  $x$ .

$x = \dots\dots\dots$  [1]

- (ii) State the name of the particle represented by the symbol  ${}_{-1}^0\text{e}$ .

$\dots\dots\dots$  [1]

- (b) Some data for the nuclei in the reaction are given in Fig. 8.1.

|               |                              | mass/u  | binding energy per nucleon /MeV |
|---------------|------------------------------|---------|---------------------------------|
| uranium-235   | ( ${}_{92}^{235}\text{U}$ )  | 235.123 |                                 |
| molybdenum-95 | ( ${}_{42}^{95}\text{Mo}$ )  | 94.945  | 8.09                            |
| lanthanum-139 | ( ${}_{57}^{139}\text{La}$ ) | 138.955 | 7.92                            |
| proton        | ( ${}_1^1\text{p}$ )         | 1.007   |                                 |
| neutron       | ( ${}_0^1\text{n}$ )         | 1.009   |                                 |

Fig. 8.1

Use data from Fig. 8.1 to

- (i) determine the binding energy, in u, of a nucleus of uranium-235,

binding energy =  $\dots\dots\dots$  u [3]

(ii) show that the binding energy per nucleon of a nucleus of uranium-235 is 7.18 MeV.

[3]

- (c) The kinetic energy of the neutron before the reaction is negligible.  
Use data from (b) to calculate the total energy, in MeV, released in this reaction.

energy = ..... MeV [2]



14. M/J/13/41/Q8

(a) State what is meant by a *nuclear fusion reaction*.

.....  
 .....  
 ..... [2]

(b) One nuclear reaction that takes place in the core of the Sun is represented by the equation



Data for the nuclei are given in Fig. 8.1.

|                            | mass/u  |
|----------------------------|---------|
| proton ${}^1_1\text{H}$    | 1.00728 |
| deuterium ${}^2_1\text{H}$ | 2.01410 |
| helium ${}^3_2\text{He}$   | 3.01605 |

**Fig. 8.1**

(i) Calculate the energy, in joules, released in this reaction.

energy = ..... J [3]

(ii) The temperature in the core of the Sun is approximately  $1.6 \times 10^7 \text{ K}$ .  
 Suggest why such a high temperature is necessary for this reaction to take place.

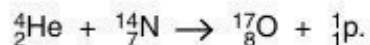
.....  
 .....  
 ..... [2]

**15. M/J/13/42/Q8**

- (a) Explain why the mass of an  $\alpha$ -particle is less than the total mass of two individual protons and two individual neutrons.

.....  
 .....  
 ..... [2]

- (b) An equation for one possible nuclear reaction is



Data for the masses of the nuclei are given in Fig. 8.1.

|             |                     | mass / u |
|-------------|---------------------|----------|
| proton      | ${}^1_1\text{p}$    | 1.00728  |
| helium-4    | ${}^4_2\text{He}$   | 4.00260  |
| nitrogen-14 | ${}^{14}_7\text{N}$ | 14.00307 |
| oxygen-17   | ${}^{17}_8\text{O}$ | 16.99913 |

**Fig. 8.1**

- (i) Calculate the mass change, in u, associated with this reaction.

mass change = ..... u [2]

- (ii) Calculate the energy, in J, associated with the mass change in (i).

energy = ..... J [2]

- (iii) Suggest and explain why, for this reaction to occur, the helium-4 nucleus must have a minimum speed.

.....

.....

..... [2]

16. O/N/13/41/Q8

(a) State what is meant by *nuclear binding energy*.

.....  
 .....  
 ..... [2]

(b) The variation with nucleon number  $A$  of the binding energy per nucleon  $B_E$  is shown in Fig. 8.1.

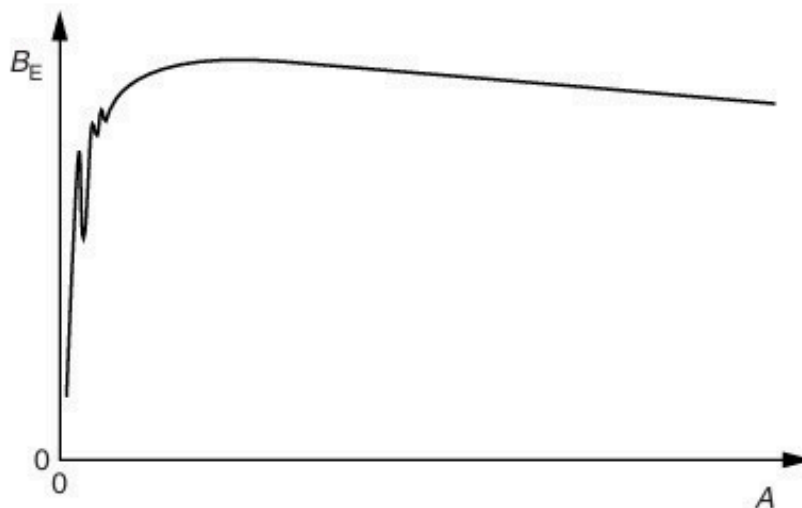
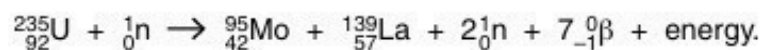


Fig. 8.1

When uranium-235 ( $^{235}_{92}\text{U}$ ) absorbs a slow-moving neutron, one possible nuclear reaction is



(i) State the name of this type of nuclear reaction.

..... [1]

(ii) On Fig. 8.1, mark the position of

1. the uranium-235 nucleus (label this position U), [1]
2. the molybdenum-95 ( $^{95}_{42}\text{Mo}$ ) nucleus (label this position Mo), [1]
3. the lanthanum-139 ( $^{139}_{57}\text{La}$ ) nucleus (label this position La). [1]

(iii) The masses of some particles and nuclei are given in Fig. 8.2.

|                   | mass/u               |
|-------------------|----------------------|
| $\beta$ -particle | $5.5 \times 10^{-4}$ |
| neutron           | 1.009                |
| proton            | 1.007                |
| uranium-235       | 235.123              |
| molybdenum-95     | 94.945               |
| lanthanum-139     | 138.955              |

**Fig. 8.2**

Calculate, for this reaction,

1. the change, in u, of the rest mass,

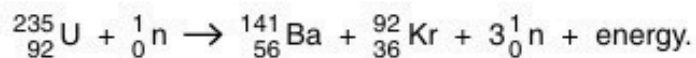
change in mass = ..... u [2]

2. the energy released, in MeV, to three significant figures.

energy = ..... MeV [3]

17. O/N/13/43/Q8

One possible nuclear fission reaction is



Barium-141 ( ${}_{56}^{141}\text{Ba}$ ) and krypton-92 ( ${}_{36}^{92}\text{Kr}$ ) are both  $\beta$ -emitters.

Barium-141 has a half-life of 18 minutes and a decay constant of  $6.4 \times 10^{-4} \text{ s}^{-1}$ .

The half-life of krypton-92 is 3.0 seconds.

(a) State what is meant by *decay constant*.

.....  
.....  
.....[2]

(b) A mass of 1.2g of uranium-235 undergoes this nuclear reaction in a very short time (a few nanoseconds).

(i) Calculate the number of barium-141 nuclei that are present immediately after the reaction has been completed.

number = ..... [2]

(ii) Using your answer in (b)(i), calculate the total activity of the barium-141 and the krypton-92 a time of 1.0 hours after the fission reaction has taken place.

activity = ..... Bq [4]

**18. M/J/14/41/Q10**

- (a) Explain what is meant by the *binding energy* of a nucleus.

.....

.....

..... [2]

- (b) Data for the masses of some particles are given in Fig. 10.1.

|                                               | mass/u    |
|-----------------------------------------------|-----------|
| proton                                        | 1.00728   |
| neutron                                       | 1.00867   |
| tritium ( ${}^3_1\text{H}$ ) nucleus          | 3.01551   |
| polonium ( ${}^{210}_{84}\text{Po}$ ) nucleus | 209.93722 |

**Fig. 10.1**

The energy equivalent of 1.0 u is 930 MeV.

- (i) Calculate the binding energy, in MeV, of a tritium ( ${}^3_1\text{H}$ ) nucleus.

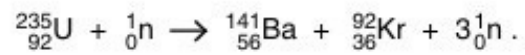
binding energy = ..... MeV [3]

- (ii) The total mass of the separate nucleons that make up a polonium-210 ( ${}^{210}_{84}\text{Po}$ ) nucleus is 211.70394 u.

Calculate the binding energy per nucleon of polonium-210.

binding energy per nucleon = ..... MeV [3]

(c) One possible fission reaction is



By reference to binding energy, explain, without any calculation, why this fission reaction is energetically possible.

.....  
.....  
..... [2]



**19. M/J/14/42/Q9**

Some water becomes contaminated with radioactive iodine-131 ( $^{131}_{53}\text{I}$ ).  
The activity of the iodine-131 in 1.0 kg of this water is 460 Bq.  
The half-life of iodine-131 is 8.1 days.

**(a)** Define radioactive *half-life*.

.....  
.....  
..... [2]

**(b) (i)** Calculate the number of iodine-131 atoms in 1.0 kg of this water.

number = ..... [3]

**(ii)** An amount of 1.0 mol of water has a mass of 18 g.

Calculate the ratio

$$\frac{\text{number of molecules of water in 1.0 kg of water}}{\text{number of atoms of iodine-131 in 1.0 kg of contaminated water}}$$

ratio = ..... [2]

- (c) An acceptable limit for the activity of iodine-131 in water has been set as  $170 \text{ Bq kg}^{-1}$ .

Calculate the time, in days, for the activity of the contaminated water to be reduced to this acceptable level.

time = ..... days [3]

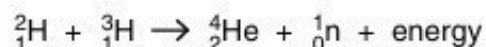
**20. O/N/14/41/Q9**

One likely means by which nuclear fusion may be achieved on a practical scale is the D-T reaction.

- (a)** State what is meant by *nuclear fusion*.

.....  
 ..... [1]

- (b)** In the D-T reaction, a deuterium ( ${}^2_1\text{H}$ ) nucleus fuses with a tritium ( ${}^3_1\text{H}$ ) nucleus to form a helium-4 ( ${}^4_2\text{He}$ ) nucleus. The nuclear equation for the reaction is



Some data for this reaction are given in Fig. 9.1.

|                                | mass/u  |
|--------------------------------|---------|
| deuterium ( ${}^2_1\text{H}$ ) | 2.01356 |
| tritium ( ${}^3_1\text{H}$ )   | 3.01551 |
| helium-4 ( ${}^4_2\text{He}$ ) | 4.00151 |
| neutron ( ${}^1_0\text{n}$ )   | 1.00867 |

**Fig. 9.1**

- (i)** Calculate the energy, in MeV, equivalent to 1.00 u. Explain your working.

energy = ..... MeV [3]

- (ii)** Use data from Fig. 9.1 and your answer in **(i)** to determine the energy released in this D-T reaction.

- (iii) Suggest why, for the D-T reaction to take place, the temperature of the deuterium and the tritium must be high.

.....

.....

..... [2]

**21. O/N/14/43/Q9**

During the de-commissioning of a nuclear reactor, a mass of  $2.5 \times 10^5 \text{ kg}$  of steel is found to be contaminated with radioactive nickel-63 ( $^{63}_{28}\text{Ni}$ ).

The total activity of the steel due to the nickel-63 contamination is  $1.7 \times 10^{14} \text{ Bq}$ .

- (a) Calculate the activity per unit mass of the steel.

activity per unit mass = .....  $\text{Bq kg}^{-1}$  [1]

- (b) Special storage precautions need to be taken when the activity per unit mass due to contamination exceeds  $400 \text{ Bq kg}^{-1}$ .

Nickel-63 is a  $\beta$ -emitter with a half-life of 92 years.

The maximum energy of an emitted  $\beta$ -particle is  $0.067 \text{ MeV}$ .

- (i) Use your answer in (a) to calculate the energy, in J, released per second in a mass of  $1.0 \text{ kg}$  of steel due to the radioactive decay of the nickel.

energy = ..... J [1]

- (ii) Use your answer in (i) to suggest, with a reason, whether the steel will be at a high temperature.

.....  
.....  
..... [1]

- (iii) Use your answer in (a) to determine the time interval before special storage precautions for the steel are not required.

time = ..... years [3]

**22. M/J/15/41/Q9**

- (a)** An isotope of an element is radioactive. Explain what is meant by *radioactive decay*.

.....

.....

.....

..... [3]

- (b)** At time  $t$ , a sample of a radioactive isotope contains  $N$  nuclei. In a short time  $\Delta t$ , the number of nuclei that decay is  $\Delta N$ .

State expressions, in terms of the symbols  $t$ ,  $\Delta t$ ,  $N$  and  $\Delta N$  for

- (i)** the number of undecayed nuclei at time  $(t + \Delta t)$ ,

number = ..... [1]

- (ii)** the mean activity of the sample during the time interval  $\Delta t$ ,

mean activity = ..... [1]

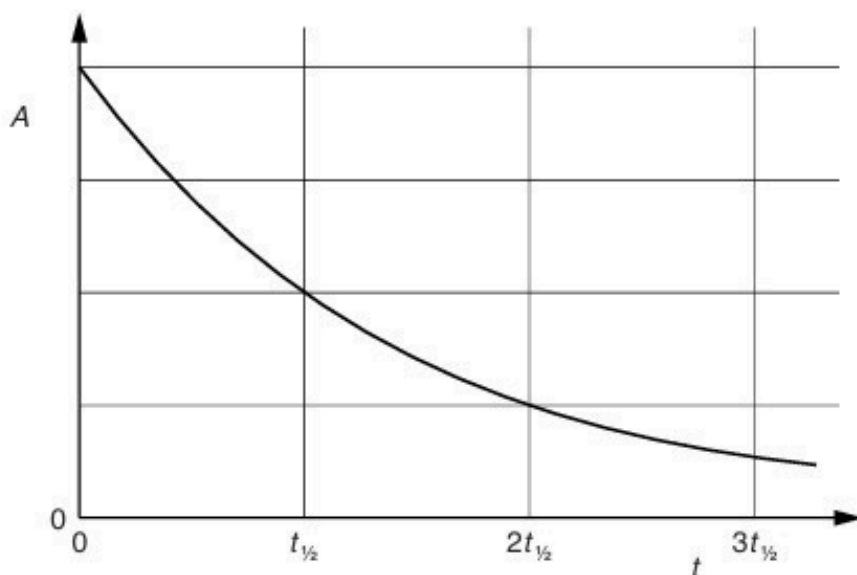
- (iii)** the probability of decay of a nucleus during the time interval  $\Delta t$ ,

probability = ..... [1]

- (iv)** the decay constant.

decay constant = ..... [1]

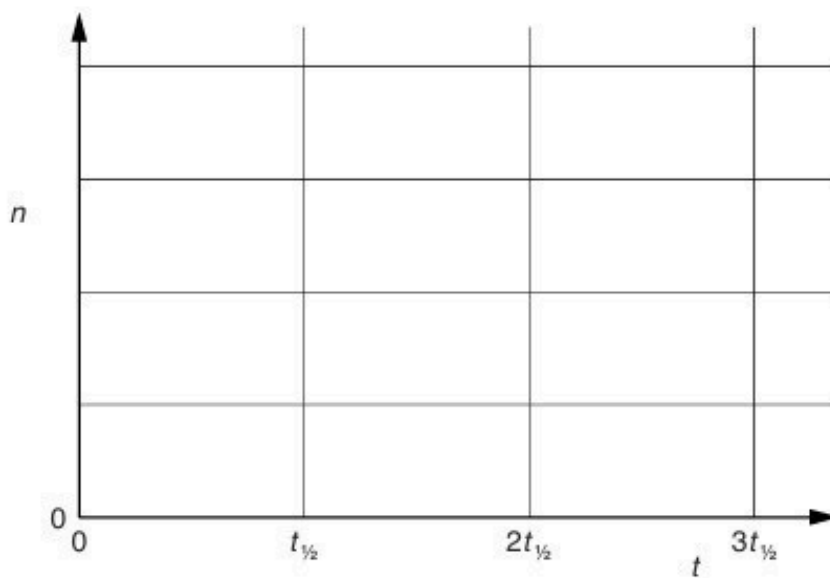
- (c) The variation with time  $t$  of the activity  $A$  of a sample of a radioactive isotope is shown in Fig. 9.1.



**Fig. 9.1**

The radioactive isotope decays to form a stable isotope S. At time  $t = 0$ , there are no nuclei of S in the sample.

On the axes of Fig. 9.2, sketch a graph to show the variation with time  $t$  of the number  $n$  of nuclei of S in the sample.



**Fig. 9.2**



**23. M/J/15/42/Q8**

The power for a space probe is to be supplied by the energy released when plutonium-236 decays by the emission of  $\alpha$ -particles.

The  $\alpha$ -particles, each of energy 5.75 MeV, are captured and their energy is converted into electrical energy with an efficiency of 24%.

**(a)** Calculate

**(i)** the energy, in joules, equal to 5.75 MeV,

energy = ..... J [1]

**(ii)** the number of  $\alpha$ -particles per second required to generate 1.9 kW of electrical power.

number per second = ..... s<sup>-1</sup> [2]

**(b)** Each plutonium-236 nucleus, on disintegration, produces one  $\alpha$ -particle.  
Plutonium-236 has a half-life of 2.8 years.

**(i)** Calculate the decay constant, in s<sup>-1</sup>, of plutonium-236.

decay constant = ..... s<sup>-1</sup> [2]

- (ii) Use your answers in (a)(ii) and (b)(i) to determine the mass of plutonium-236 required for the generation of 1.9 kW of electrical power.

mass = ..... g [4]

- (c) The minimum electrical power required for the space probe is 0.84 kW.

Calculate the time, in years, for which the sample of plutonium-236 in (b)(ii) will provide sufficient power.

time = ..... years [2]

**24. O/N/15/41/Q8**

- (a)** Distinguish, for an atom, between a nucleus and a nucleon.

nucleus: .....

.....

nucleon: .....

.....

[3]

- (b)** Radon gas is a naturally occurring radioactive gas with a half-life of 3.8 days.

The activity of radon gas in a room is found to be 97 Bq in each  $1.0\text{ m}^3$  of air.

- (i)** Calculate

1. the decay constant, in  $\text{s}^{-1}$ , of radon,

decay constant = .....  $\text{s}^{-1}$  [2]

2. the number of radon atoms giving rise to an activity of 97 Bq.

number = ..... [2]

- (ii) A volume of  $2.5 \times 10^{-2} \text{ m}^3$  of air in the room contains 1.0 mol of molecules.

Determine the ratio, for  $1.0 \text{ m}^3$  of air,

$$\frac{\text{number of radon atoms}}{\text{number of air molecules}} .$$

ratio = ..... [2]

**25. O/N/15/43/Q9**

**(a)** State what is meant by the *binding energy* of a nucleus.

.....

.....

..... [2]

**(b)** Data for two isotopes of uranium are given in Fig. 9.1.

| isotope     | binding energy per nucleon/MeV | binding energy/MeV |
|-------------|--------------------------------|--------------------|
| uranium-235 | 7.59                           | .....              |
| uranium-238 | .....                          | 1802               |

**Fig. 9.1**

**(i)** State what is meant by *isotopes*.

.....

.....

..... [2]

**(ii)** Complete Fig. 9.1.

[2]

**(c)** Uranium-235 has a half-life of  $7.1 \times 10^8$  years.

**(i)** Show that the decay constant  $\lambda$  of uranium-235 is  $3.1 \times 10^{-17} \text{ s}^{-1}$ .

[1]

(ii) A sample of uranium-235 has an activity of  $5.0 \times 10^3 \text{ Bq}$ .

Calculate the mass of the sample.

mass = ..... g [3]

**26. F/M/16/42/Q13**

Beryllium-7 ( ${}^7_4\text{Be}$ ) is produced in the upper atmosphere and then sinks down onto the Earth's surface. Nuclei of beryllium-7 decay with a half-life of 53.3 days to form stable nuclei.

The activity of a sample of beryllium-7 on a tree leaf is 39 mBq.

**(a)** Show that the decay constant of beryllium-7 is  $1.5 \times 10^{-7} \text{ s}^{-1}$ .

[1]

**(b)** Determine the mass of the beryllium-7 on the leaf.

mass = ..... kg [3]

**(c)** The leaf is covered so that no further beryllium-7 is added to the existing sample from the atmosphere.

Calculate the time that must elapse before the activity of the sample is reduced to 2.0 mBq.

time = ..... s [2]

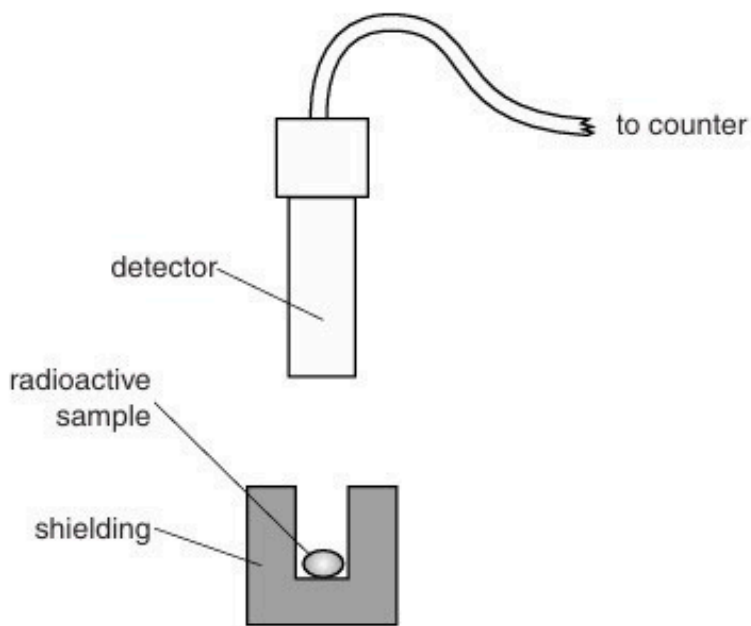
[Total: 6]

**27. M/J/16/41/Q13**

Copper-66 is a radioactive isotope.

When a nucleus of copper-66 decays, the emissions include a  $\beta^-$  particle and a  $\gamma$ -ray photon.

The count rate produced from a sample of the isotope copper-66 is measured using a detector and counter, as illustrated in Fig. 13.1.



**Fig. 13.1**

- (a) State three reasons why the activity of the sample of copper-66 is not equal to the measured count rate.

1. ....
2. ....
3. ....



- (b) In a time of 42.0 minutes, the count rate from the sample of copper-66 is found to decrease from  $3.62 \times 10^4$  Bq to  $1.21 \times 10^2$  Bq.

Calculate the half-life of copper-66.

half-life = ..... minutes [2]

- (c) The  $\gamma$ -ray photons emitted from radioactive nuclei have specific energies, dependent on the nucleus emitting the photons.

By comparison with emission line spectra, suggest what can be deduced about energy levels in nuclei.

.....  
..... [1]

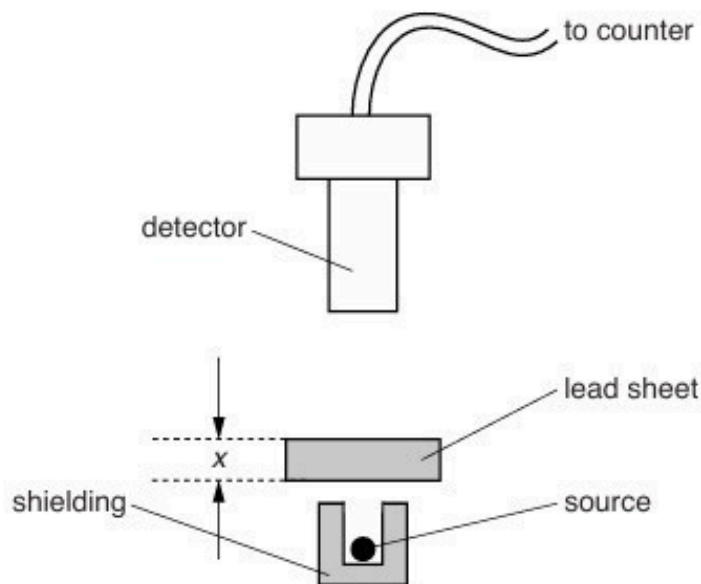
[Total: 6]

**28. M/J/16/42/Q13**

- (a)** Explain what is meant by *gamma radiation* ( $\gamma$ -radiation).

.....  
.....  
..... [2]

- (b)** A source of gamma radiation is placed a fixed distance away from a detector and counter, as illustrated in Fig. 13.1.

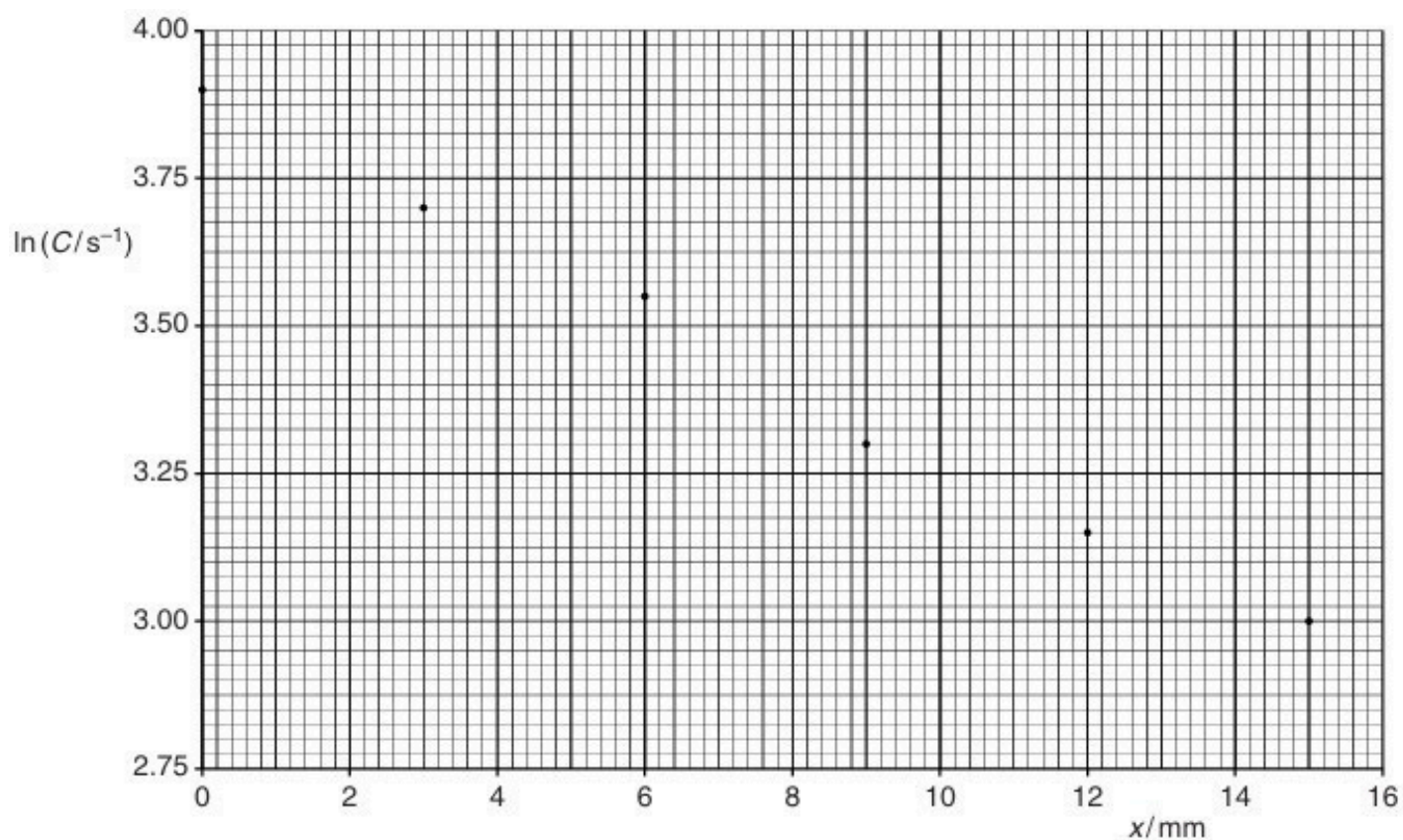


**Fig. 13.1**

A sheet of lead of thickness  $x$  is placed between the source and the detector.

The average count rate  $C$ , corrected for background, is recorded. This is repeated for different values of  $x$ .

The variation with thickness  $x$  of  $\ln C$  is shown in Fig. 13.2.



**Fig. 13.2**

The absorption of gamma radiation in lead may be represented by the equation

$$C = C_0 e^{-\mu x}$$

where  $C_0$  is the count rate for  $x = 0$  and  $\mu$  is the linear attenuation (absorption) coefficient.

Use Fig. 13.2 to determine the linear attenuation coefficient  $\mu$  for this gamma radiation in lead.

$\mu = \dots\dots\dots \text{mm}^{-1}$  [4]

**(c)** The value of  $\mu$  calculated in **(b)** is for gamma radiation in lead.

Suggest and explain whether the value of  $\mu$  for aluminium would be the same, greater or smaller.

.....

.....

..... [2]

[Total: 8]

**29. O/N/16/41/Q12**

Radon-222 ( $^{222}_{86}\text{Rn}$ ) is a radioactive element found in atmospheric air.  
The decay constant of radon-222 is  $2.1 \times 10^{-6} \text{ s}^{-1}$ .

- (a) (i) Define radioactive *half-life*.

.....  
.....  
..... [2]

- (ii) Show that the half-life  $t_{\frac{1}{2}}$  is related to the decay constant  $\lambda$  by the expression

$$\lambda t_{\frac{1}{2}} = 0.693.$$

[2]

- (b) Radon-222 is considered to be an unacceptable health hazard when the activity of radon-222 is greater than 200 Bq in  $1.0 \text{ m}^3$  of air.

Calculate the minimum mass of radon-222 in  $1.0 \text{ m}^3$  of air above which the health hazard becomes unacceptable.

mass = ..... kg [4]

[Total: 8]

**30. O/N/16/42/Q14**

Phosphorus-30 ( $^{30}_{15}\text{P}$ ) was the first artificial radioactive nuclide to be produced in a laboratory. This was achieved by bombarding aluminium-27 ( $^{27}_{13}\text{Al}$ ) with  $\alpha$ -particles.

A partial nuclear equation to represent this reaction is



**(a)** State the full nuclear notation for

**(i)** the  $\alpha$ -particle,

.....[1]

**(ii)** the particle represented by the symbol  $\Phi$ .

.....[1]

- (b) Data for the rest masses of the particles in the reaction are given in Fig. 14.1.

| particle                | mass / u |
|-------------------------|----------|
| ${}^{27}_{13}\text{Al}$ | 26.98153 |
| $\alpha$                | 4.00260  |
| ${}^{30}_{15}\text{P}$  | 29.97830 |
| $\Phi$                  | 1.00867  |

**Fig. 14.1**

Calculate, for this reaction,

- (i) the change in the total rest mass of the particles,

mass change = ..... u [2]

- (ii) the energy, in joule, equivalent to the mass change calculated in (i).

energy = ..... J [2]

- (c) With reference to your answer in (b)(i), comment on the energy of the  $\alpha$ -particle such that the reaction can take place.

.....  
 .....  
 .....  
 ..... [2]

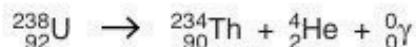
[Total: 8]

**31. F/M/17/42/Q12**

- (a) Define the *binding energy* of a nucleus.

.....  
 .....  
 .....[2]

- (b) A stationary nucleus of uranium-238 ( $^{238}_{92}\text{U}$ ) decays to form a nucleus of thorium-234 ( $^{234}_{90}\text{Th}$ ). An  $\alpha$ -particle and a gamma-ray photon are emitted. The equation representing the decay is



The masses of the nuclei are given in Fig. 12.1.

| nucleus     | mass/u    |
|-------------|-----------|
| uranium-238 | 238.05076 |
| thorium-234 | 234.04357 |
| helium-4    | 4.00260   |

**Fig. 12.1**

- (i) State the relationship between the binding energies of the nuclei that is consistent with this reaction being energetically possible.

.....  
 .....[1]

- (ii) Calculate, for this reaction,

1. the change, in u, of the mass,

change of mass = .....u [1]

2. the total energy, in J, released.

energy = .....J [2]



- (iii) State and explain whether the energy of the gamma-ray photon is equal to the energy released in the reaction.

.....

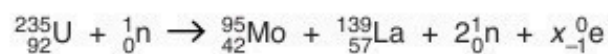
.....

.....[2]

[Total: 8]

**32. M/J/17/41/Q12**

One possible nuclear reaction that takes place in a nuclear reactor is given by the equation



Data for the nuclei and particles are given in Fig. 12.1.

| nucleus or particle      | mass/u                |
|--------------------------|-----------------------|
| ${}_{92}^{235}\text{U}$  | 235.123               |
| ${}_{42}^{95}\text{Mo}$  | 94.945                |
| ${}_{57}^{139}\text{La}$ | 138.955               |
| ${}_0^1\text{n}$         | 1.00863               |
| ${}_{-1}^0\text{e}$      | $5.49 \times 10^{-4}$ |

**Fig. 12.1**

**(a)** Determine, for this nuclear reaction, the value of  $x$ .

$x = \dots\dots\dots[1]$

**(b) (i)** Show that the energy equivalent to 1.00 u is 934 MeV.

[3]

**(ii)** Calculate the energy, in MeV, released in this reaction. Give your answer to three significant figures.

energy = ..... MeV [3]

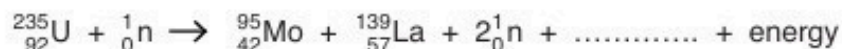
**(c)** Suggest the forms of energy into which the energy calculated in **(b)(ii)** is transformed.

.....  
.....  
.....  
.....[2]

[Total: 9]

**33. M/J/17/42/Q12**

One nuclear reaction that can take place in a nuclear reactor may be represented, in part, by the equation



Data for a nucleus and some particles are given in Fig. 12.1.

| nucleus or particle      | mass/u                |
|--------------------------|-----------------------|
| ${}_{57}^{139}\text{La}$ | 138.955               |
| ${}_0^1\text{n}$         | 1.00863               |
| ${}_1^1\text{p}$         | 1.00728               |
| ${}_{-1}^0\text{e}$      | $5.49 \times 10^{-4}$ |

**Fig. 12.1**

(a) Complete the nuclear reaction shown above. [1]

(b) (i) Show that the energy equivalent to 1.00 u is 934 MeV.

[3]

(ii) Calculate the binding energy per nucleon, in MeV, of lanthanum-139 ( ${}_{57}^{139}\text{La}$ ).

binding energy per nucleon = ..... MeV [3]

- (c) State and explain whether the binding energy per nucleon of uranium-235 ( $^{235}_{92}\text{U}$ ) will be greater, equal to or less than your answer in (b)(ii).

.....

.....

.....

.....[3]

[Total: 10]

34. O/N/17/41/Q12

- (a) A radiation detector is placed close to a radioactive source. The detector does not surround the source.  
Radiation is emitted in all directions and, as a result, the activity of the source and the measured count rate are different.

Suggest two other reasons why the activity and the measured count rate may be different.

1. ....

.....

2. ....

.....

[2]

- (b) The variation with time  $t$  of the measured count rate in (a) is shown in Fig. 12.1.

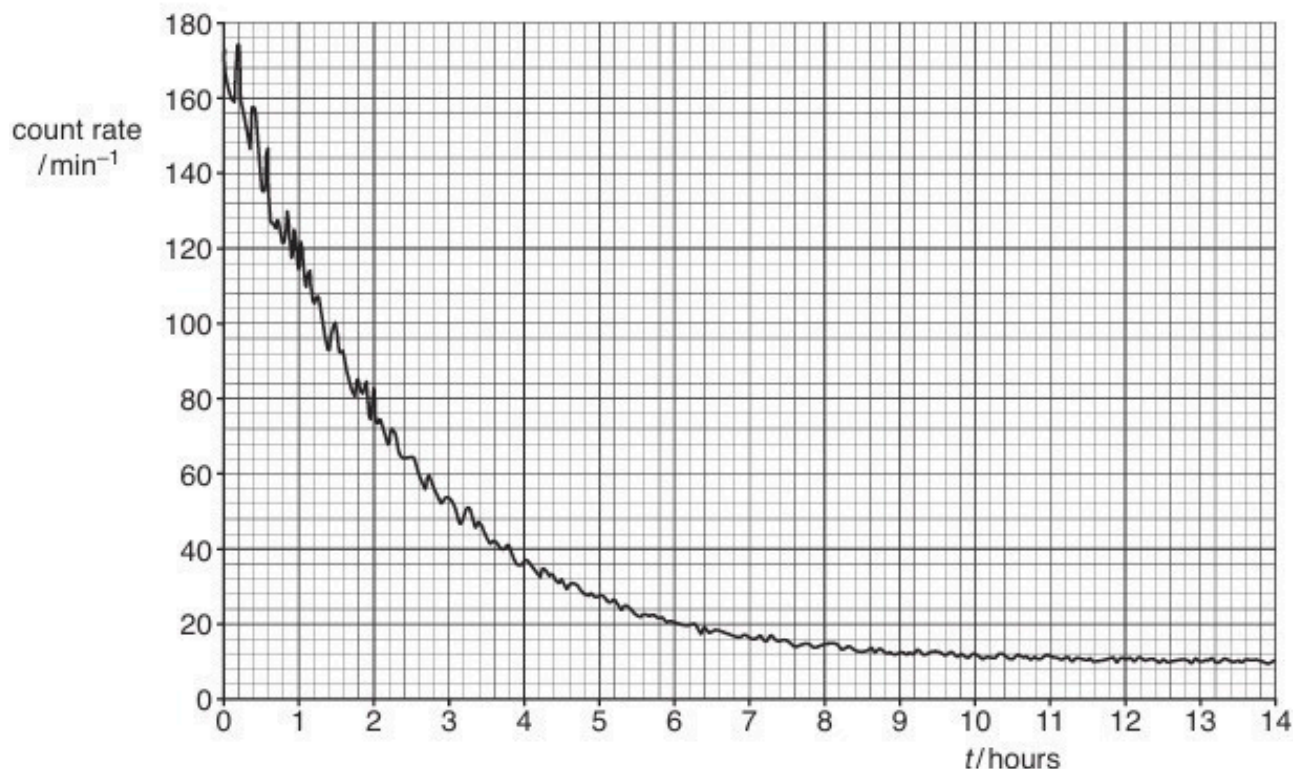


Fig. 12.1

- (i) State the feature of Fig. 12.1 that indicates the random nature of radioactive decay.

.....

..... [1]

(ii) Use Fig. 12.1 to determine the half-life of the radioactive isotope in the source.

half-life = ..... hours [4]

- (c) The readings in (b) were obtained at room temperature.  
A second sample of this isotope is heated to a temperature of 500 °C.  
The initial count rate at time  $t = 0$  is the same as that in (b).  
The variation with time  $t$  of the measured count rate from the heated source is determined.

State, with a reason, the difference, if any, in

1. the half-life,

.....  
.....  
.....

2. the measured count rate for any specific time.

.....  
.....  
.....

[3]

[Total: 10]

35. O/N/17/42/Q12

The isotope iodine-131 ( $^{131}_{53}\text{I}$ ) is radioactive with a decay constant of  $8.6 \times 10^{-2} \text{ day}^{-1}$ .  
 $\beta^-$  particles are emitted with a maximum energy of 0.61 MeV.

(a) State what is meant by

(i) *radioactive*,

.....  
.....  
.....[2]

(ii) *decay constant*.

.....  
.....  
.....[2]

(b) Explain why the emitted  $\beta^-$  particles have a range of energies.

.....  
.....  
.....[2]

(c) A sample of blood contains  $1.2 \times 10^{-9} \text{ g}$  of iodine-131.

Determine, for this sample of blood,

(i) the activity of the iodine-131,

activity = ..... Bq [3]



- (ii) the time for the activity of the iodine-131 to be reduced to 1/50 of the activity calculated in (i).

time = ..... days [2]

[Total: 11]

36. F/M/18/42/Q13

- (a) (i) Define radioactive *decay constant*.

.....  
.....  
.....[2]

- (ii) Show that the decay constant  $\lambda$  is related to the half-life  $t_{\frac{1}{2}}$  of a radioactive isotope by the expression

$$\lambda t_{\frac{1}{2}} = \ln 2$$

[2]

- (b) A small volume of solution containing the radioactive isotope sodium-24 ( ${}^{24}_{11}\text{Na}$ ) has an initial activity of  $3.8 \times 10^4 \text{ Bq}$ . Sodium-24, of half-life 15 hours, decays to form a stable daughter isotope.

All of the solution is poured into a container of water. After 36 hours, a sample of water of volume  $5.0 \text{ cm}^3$ , taken from the container, is found to have an activity of  $1.2 \text{ Bq}$ .

Assuming that the solution of the radioactive isotope is distributed uniformly throughout the container of water, calculate the volume of water in the container.

volume = .....  $\text{cm}^3$  [4]

[Total: 8]

**37. M/J/18/41/Q13**

(a) State what is meant by *radioactive decay*.

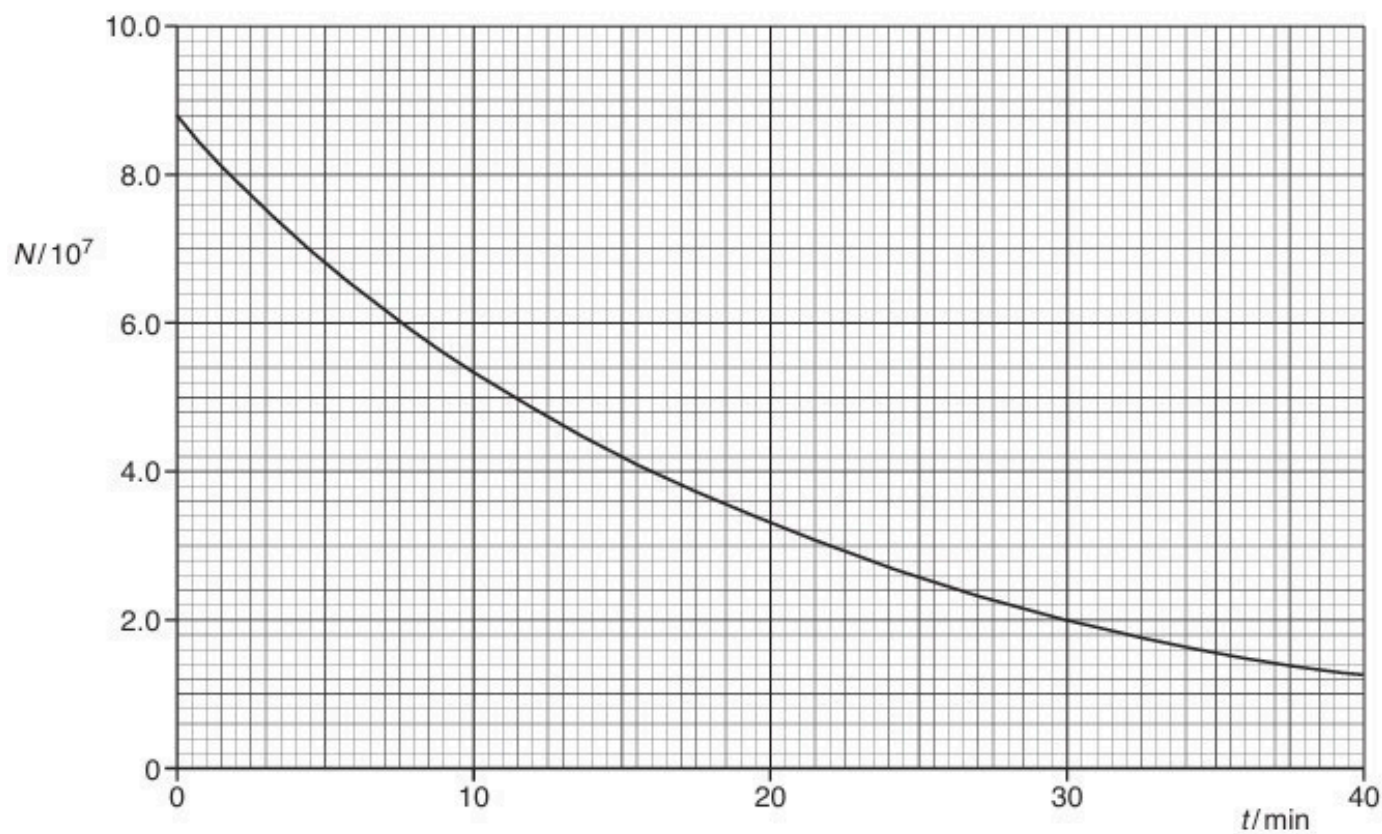
.....

.....

.....

.....[2]

(b) The variation with time  $t$  of the number  $N$  of technetium-101 nuclei in a sample of radioactive material is shown in Fig. 13.1.



**Fig. 13.1**

- (i) Use Fig. 13.1 to determine the activity, in Bq, of the sample of technetium-101 at time  $t = 14.0$  minutes. Show your working.

activity = ..... Bq [4]

- (ii) Without calculating the half-life of technetium-101, use your answer in (i) to determine the decay constant  $\lambda$  of technetium-101.

$\lambda = \dots\dots\dots \text{s}^{-1}$  [2]

[Total: 8]

38. M/J/18/42/Q12

(a) State what is meant by *radioactive decay*.

.....

.....

.....

.....[2]

(b) An unstable nuclide P has decay constant  $\lambda_P$  and decays to form a nuclide D. This nuclide D is unstable and decays with decay constant  $\lambda_D$  to form a stable nuclide S. The decay chain is illustrated in Fig. 12.1.

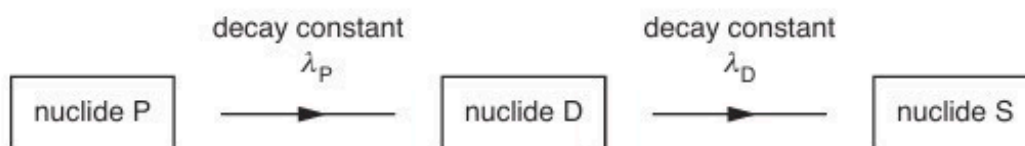
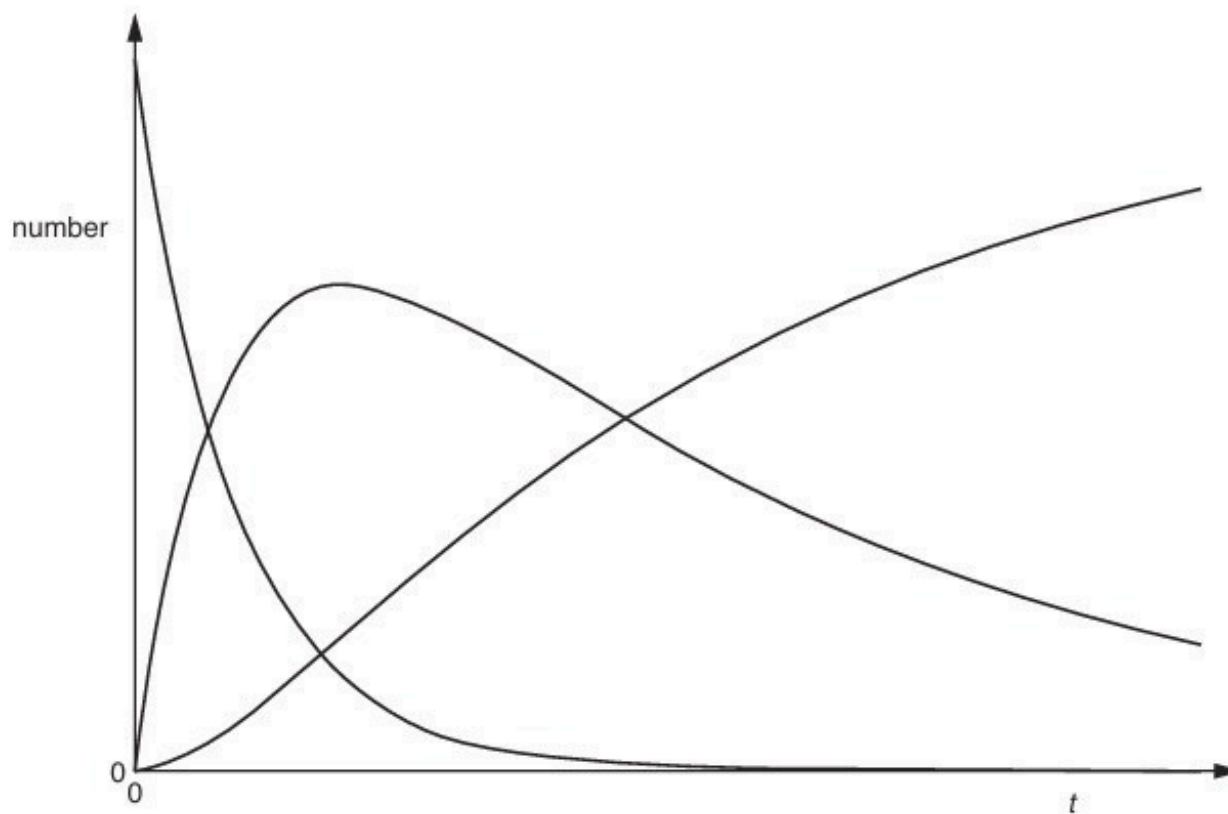


Fig. 12.1

The symbols P, D and S are not the nuclide symbols.

Initially, a radioactive sample contains only nuclide P.

The variation with time  $t$  of the number of nuclei of each of the three nuclides in the sample is shown in Fig. 12.2.



(i) On Fig. 12.2, use the symbols P, D and S to identify the curve for each of the three nuclides. [2]

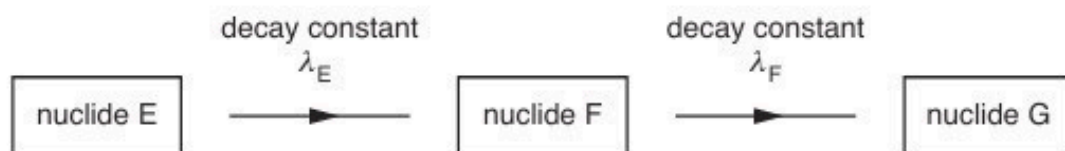
(ii) The half-life of nuclide P is 60.0 minutes.

Calculate the decay constant  $\lambda_P$  in  $\text{s}^{-1}$ , of this nuclide.

$$\lambda_P = \dots\dots\dots \text{s}^{-1} \quad [2]$$

(c) In the decay chain shown in Fig. 12.1,  $\lambda_P$  is approximately equal to  $5\lambda_D$ .

The decay chain of a different nuclide E is illustrated in Fig. 12.3.



**Fig. 12.3**

The decay constant  $\lambda_F$  of nuclide F is very much larger than the decay constant  $\lambda_E$  of nuclide E.

By reference to the half-life of nuclide F, explain why the number of nuclei of nuclide F in the sample is always small.

.....  
 .....  
 ..... [2]

[Total: 8]

39. O/N/18/41/Q12

(a) State what is meant by *radioactive decay*.

.....

.....

.....

.....[3]

(b) The variation with time  $t$  of the number  $N$  of undecayed nuclei in a sample of a radioactive isotope is shown in Fig. 12.1.

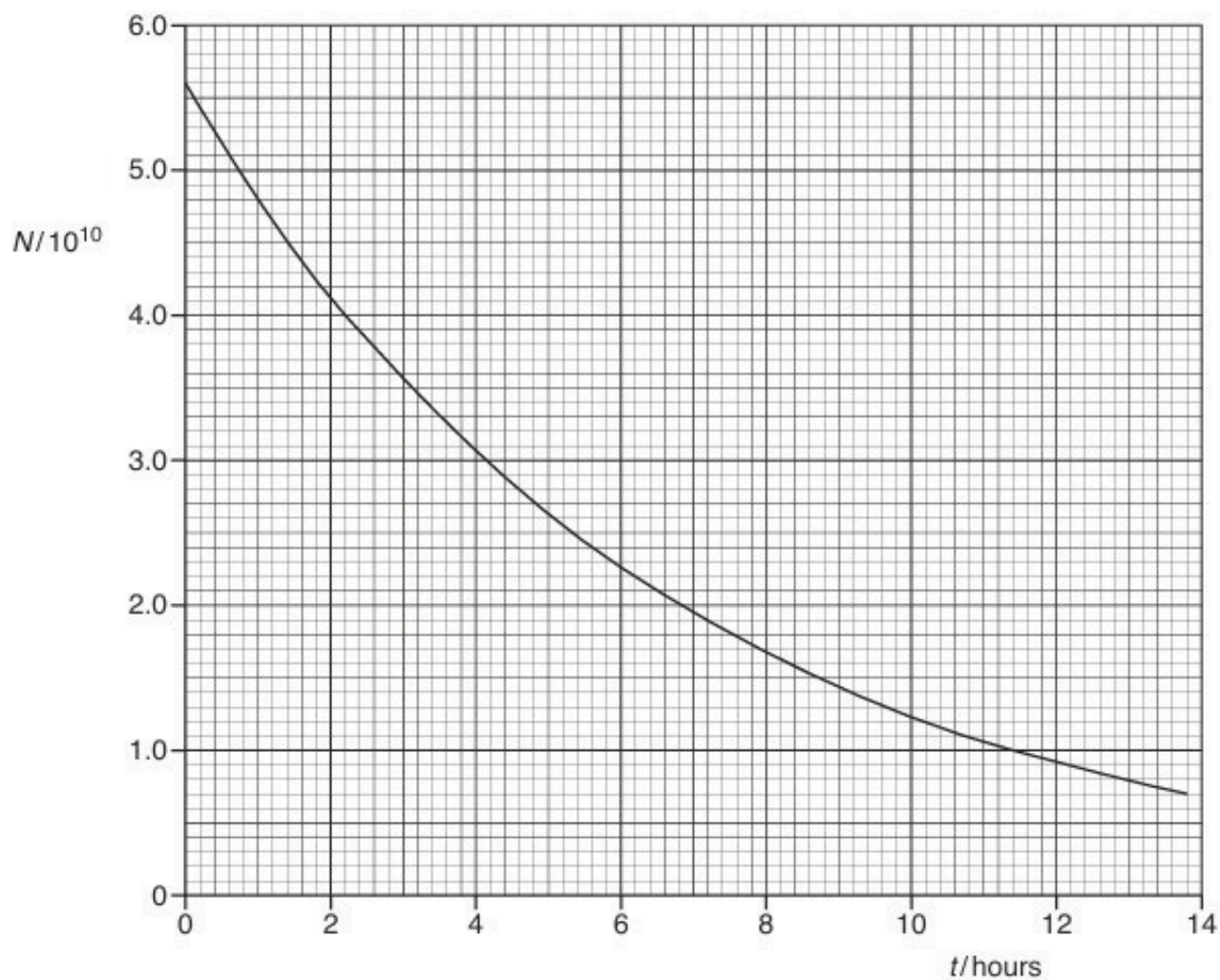


Fig. 12.1



- (i) Use the gradient of the line in Fig. 12.1 to determine the activity, in Bq, of the sample at time  $t = 4.0$  hours. Show your working.

activity = ..... Bq [3]

- (ii) Use your answer in (i) to show that the decay constant  $\lambda$  of the isotope is approximately  $4 \times 10^{-5} \text{ s}^{-1}$ .

[2]

- (c) A sample of a different radioactive isotope has an initial activity of  $4.6 \times 10^3 \text{ Bq}$ . The sample must be stored safely until its activity is reduced to  $1.0 \times 10^3 \text{ Bq}$ .

The decay constant of the isotope is  $5.5 \times 10^{-7} \text{ s}^{-1}$ . The decay products are not radioactive.

Calculate the minimum time, in days, for which the sample must be stored.

time = ..... days [3]

[Total: 11]



**40. O/N/18/422/Q12**

**(a)** State what is meant by *nuclear fusion* and *nuclear fission*.

nuclear fusion: .....

.....

.....

nuclear fission: .....

.....

.....

[3]

(b) A nuclear reaction which may, in the future, be used for the generation of electrical energy is



(i) Name the particle x.

.....[1]

(ii) Data for the binding energy per nucleon  $E_B$  of some nuclei are given in Fig. 12.1.

|           |                   | binding energy per nucleon<br>$E_B/10^{-13}\text{ J}$ |
|-----------|-------------------|-------------------------------------------------------|
| deuterium | ${}^2_1\text{H}$  | 1.7813                                                |
| tritium   | ${}^3_1\text{H}$  | 4.5285                                                |
| helium    | ${}^4_2\text{He}$ | 11.3290                                               |

**Fig. 12.1**

1. State the binding energy per nucleon of x.

binding energy per nucleon = ..... J

2. Calculate the energy change that takes place in this reaction.

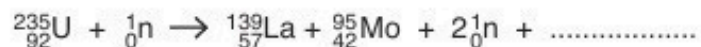
energy change = ..... J  
[3]

(iii) Use your answer in (ii) **part 2** to determine the energy release when 2.0 g of deuterium ( ${}^2_1\text{H}$ ) reacts with 3.0 g of tritium ( ${}^3_1\text{H}$ ).

energy = ..... J [1]

**41. F/M/19/42/Q12**

The incomplete nuclear equation for one possible reaction that takes place in the core of a nuclear reactor is



- (a) (i) State the name given to this type of nuclear reaction.

..... [1]

- (ii) Complete the nuclear equation.

[2]

- (b) The mass defect for the reaction is 0.223 u.

- (i) Calculate the energy, in J, equivalent to 0.223 u.

energy = ..... J [2]

- (ii) Suggest **two** forms of the energy released in this reaction.

1. ....

.....

2. ....

..... [2]

[Total: 7]

**42. M/J/19/41/Q12**

- (a) A sample of a radioactive isotope contains  $N$  nuclei of the isotope at time  $T$ .  
At time  $(T + \Delta T)$ , the sample contains  $(N - \Delta N)$  nuclei of the isotope. The time interval  $\Delta T$  is short.

Use the symbols  $N$ ,  $\Delta N$ ,  $T$  and  $\Delta T$  to give expressions for:

- (i) the average activity of the sample during the time  $\Delta T$

..... [1]

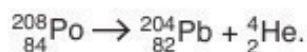
- (ii) the probability of decay of a nucleus in the time  $\Delta T$

..... [1]

- (iii) the decay constant  $\lambda$  of the isotope.

..... [1]

- (b) The isotope polonium-208 ( $^{208}_{84}\text{Po}$ ) is radioactive and decays to form lead-204 ( $^{204}_{82}\text{Pb}$ ).  
The nuclear equation for this decay is



Data for nuclear masses are given in Fig. 12.1.

|                        | mass / u    |
|------------------------|-------------|
| $^4_2\text{He}$        | 4.002 603   |
| $^{204}_{82}\text{Pb}$ | 203.973 043 |
| $^{208}_{84}\text{Po}$ | 207.981 245 |

**Fig. 12.1**

- (i) Determine, for the decay of one nucleus of polonium-208:

1. the change, in u, of the mass

mass change = ..... u [1]

2. the total energy, in pJ, released.

energy = ..... pJ [3]

(ii) The polonium-208 nucleus is initially stationary. The initial kinetic energy of the  ${}^4_2\text{He}$  nucleus ( $\alpha$ -particle) is found to be less than the energy calculated in (i) **part 2**.

Suggest **two** possible reasons for this difference.

1. ....

.....

2. ....

.....

[2]

[Total: 9]

**43. M/J/19/42/Q12**

(a) State what is meant by the *binding energy* of a nucleus.

.....

.....

.....[2]

(b) Some masses are shown in Fig. 12.1.

|                                                    | mass/u  |
|----------------------------------------------------|---------|
| proton ( ${}^1_1\text{p}$ )                        | 1.007   |
| neutron ( ${}^1_0\text{n}$ )                       | 1.009   |
| lanthanum-141 ( ${}^{141}_{57}\text{La}$ ) nucleus | 140.911 |

**Fig. 12.1**

Calculate the binding energy of a nucleus of lanthanum-141.

binding energy = ..... J [4]

(c) The nuclide lanthanum-141 ( ${}^{141}_{57}\text{La}$ ) has a half-life of 3.9 hours.

Initially, a radioactive source contains only lanthanum-141. The initial activity of the source is  $A_0$ .

(i) Calculate the time for the activity of the lanthanum-141 to be reduced to  $0.40A_0$ .

- (ii) Suggest why the total activity of the radioactive source measured at the time calculated in (i) may be greater than  $0.40A_0$ .

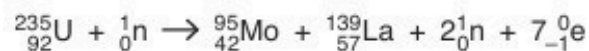
.....

.....[1]

[Total: 10]

**44. O/N/19/41/Q12**

One possible nuclear reaction that takes place is



Data for nuclei in this reaction are given in Fig. 12.1.

| nucleus                  | mass/u  | total mass of separate nucleons/u | mass defect/u | binding energy per nucleon/MeV |
|--------------------------|---------|-----------------------------------|---------------|--------------------------------|
| ${}_{42}^{95}\text{Mo}$  | 94.906  | 95.765                            | 0.859         | 8.443                          |
| ${}_{57}^{139}\text{La}$ | 138.906 | 140.125                           | 1.219         | 8.189                          |
| ${}_{92}^{235}\text{U}$  | 235.044 | 236.909                           | 1.865         | .....                          |

**Fig. 12.1**

**(a)** Show that the energy equivalent to a mass of 1.00 u is 934 MeV.

[2]

**(b) (i)** Use data from Fig. 12.1 to calculate the binding energy per nucleon of a nucleus of uranium-235 ( ${}_{92}^{235}\text{U}$ ). Complete Fig. 12.1.

[2]

**(ii)** The nucleon number of an isotope of the element rutherfordium is 267.

State whether the binding energy per nucleon of this isotope will be greater than, equal to or less than the binding energy per nucleon of uranium-235.

..... [1]



(c) Calculate the total energy, in MeV, released in this nuclear reaction.

energy = ..... MeV [2]

(d) The nuclei in  $1.2 \times 10^{-7}$  mol of uranium-235 all undergo this reaction in a time of 25 ms.

Calculate the average power release during the time of 25 ms.

power = ..... W [3]

[Total: 10]

**45. O/N/19/42/Q12**

Radon-222 ( $^{222}_{86}\text{Ra}$ ) is a radioactive gas that decays randomly with a decay constant of  $7.55 \times 10^{-3} \text{ hour}^{-1}$ .

**(a)** State what is meant by:

**(i)** *random decay*

.....  
..... [1]

**(ii)** *decay constant.*

.....  
.....  
..... [2]

**(b)** The activity of radon gas in a sample of  $4.80 \times 10^{-3} \text{ m}^3$  of air taken from a building is 0.600 Bq.

There are  $2.52 \times 10^{25}$  air molecules in a volume of  $1.00 \text{ m}^3$  of air.

Calculate, for  $1.00 \text{ m}^3$  of the air, the ratio

$$\frac{\text{number of air molecules}}{\text{number of radon atoms}}.$$

ratio = ..... [5]

[Total: 8]

**46. F/M/20/42/Q12**

- (a) Explain what is meant by the *binding energy* of a nucleus.

.....  
 .....  
 ..... [2]

- (b) The following nuclear reaction takes place:



- (i) Determine the values of  $x$  and  $y$ .

$x =$  .....

$y =$  .....

[1]

- (ii) State the name of this type of nuclear reaction.

..... [1]

- (iii) Compare the binding energy per nucleon of uranium-235 with the binding energy per nucleon of caesium-144.

.....  
 ..... [1]

- (c) Yttrium-90 decays into zirconium-90, a stable isotope.

A sample initially consists of pure yttrium-90.

Calculate the time, in days, when the ratio of the number of yttrium-90 nuclei to the number of zirconium-90 nuclei would be 2.0.

The half-life of yttrium-90 is 2.7 days.

time = ..... days [3]

**47. M/J/20/41/Q12**

- (a) The decay of a sample of a radioactive isotope is said to be random and spontaneous.

Explain what is meant by the decay being:

- (i) *random*

.....  
..... [1]

- (ii) *spontaneous*.

.....  
..... [1]

- (b) A radioactive isotope X has a half-life of 1.4 hours.

Initially, a pure sample of this isotope X has an activity of  $3.6 \times 10^5$  Bq.

Determine the activity of the isotope X in the sample after a time of 2.0 hours.

activity = ..... Bq [3]

(c) The variation with time  $t$  of the actual activity  $A$  of the sample in (b) is shown in Fig. 12.1.

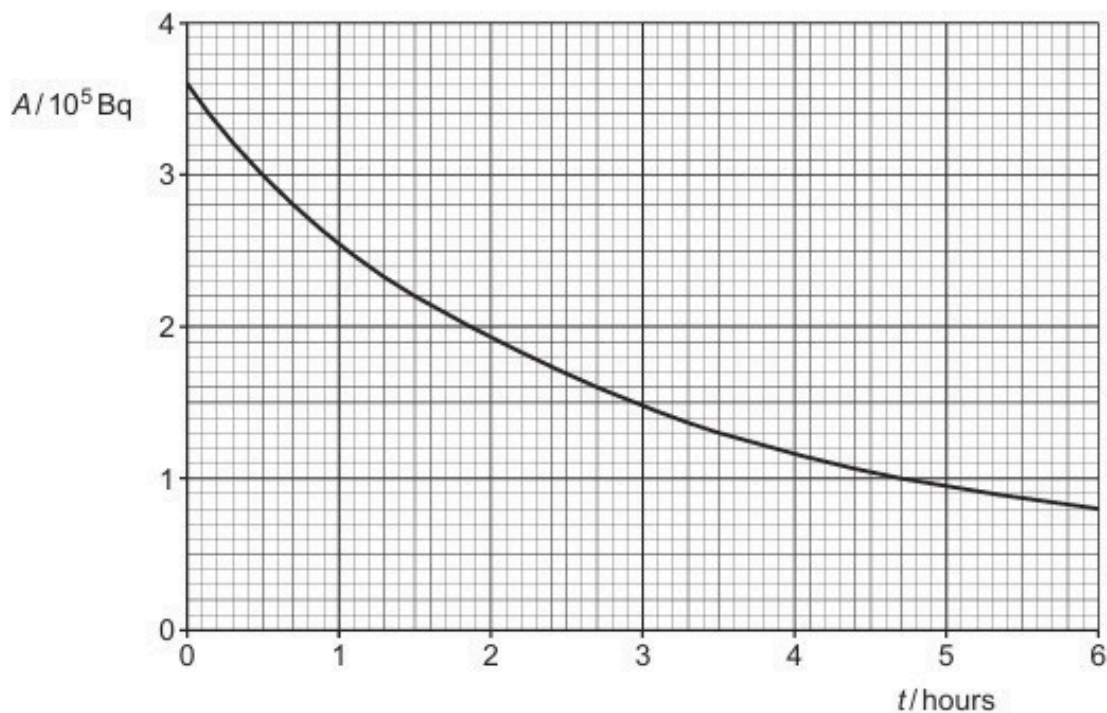


Fig. 12.1

- (i) The initial activity of isotope X in the sample is  $3.6 \times 10^5 \text{ Bq}$ .

Use information from (b) to sketch, on the axes of Fig. 12.1, the variation with time  $t$  of the activity of a pure sample of isotope X. [1]

- (ii) Suggest an explanation for any difference between the actual activity of the sample shown in Fig. 12.1 and the curve you have drawn for the activity of isotope X.

.....  
.....  
..... [2]

[Total: 8]

**48. M/J/20/42/Q12**

(a) State what is meant by the *mass defect* of a nucleus.

.....

.....

..... [2]

(b) Some masses are shown in Table 12.1.

**Table 12.1**

|                                        | mass/u    |
|----------------------------------------|-----------|
| proton ${}^1_1\text{p}$                | 1.007 276 |
| neutron ${}^1_0\text{n}$               | 1.008 665 |
| helium-4 ( ${}^4_2\text{He}$ ) nucleus | 4.001 506 |

Show that:

(i) the energy equivalence of 1.00 u is 934 MeV

[2]

(ii) the binding energy per nucleon of a helium-4 nucleus is 7.09 MeV.

[2]

- (c) Isotopes of hydrogen have binding energies per nucleon of less than 3 MeV.

Suggest why a nucleus of helium-4 does not spontaneously break down to become nuclei of hydrogen.

.....  
.....  
..... [2]

[Total: 8]

**49. O/N/20/41/Q12**

Iodine-131 ( $^{131}_{53}\text{I}$ ) is a radioactive isotope with a decay constant of  $9.9 \times 10^{-7} \text{ s}^{-1}$ .

(a) State what is meant by:

(i) *radioactive*

.....  
.....  
..... [2]

(ii) *decay constant.*

.....  
.....  
..... [2]

(b) Some water becomes contaminated with iodine-131.  
The activity of the iodine-131 in 1.0 kg of water is 560 Bq.

Determine the number of iodine-131 atoms in 1.0 kg of water.

number = ..... [2]

(c) Regulations require that the activity of iodine-131 in 1.0 kg of water is to be less than 170 Bq.

Calculate the time, in days, for the activity of the contaminated water in (b) to be reduced to 170 Bq.

time = ..... days [3]



50. O/N/20/42/Q12

(a) (i) Define nuclear *binding energy*.

.....

.....

..... [2]

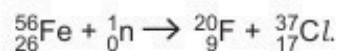
(ii) Explain what is meant by a *nuclear fission* reaction.

.....

.....

..... [2]

(b) A student suggests that one possible nuclear reaction is



The binding energy per nucleon of a nucleus varies with the nucleon number.  
Use this variation to explain why the reaction would **not** result in an overall release of energy.

.....

.....

.....

.....

..... [3]

[Total: 7]

**51. F/M/21/42/Q12**

- (a) Radioactive decay is both spontaneous and random.

State what is meant by:

1. *spontaneous decay* .....
2. *random decay* .....

[2]

- (b) Strontium-90 ( $^{90}_{38}\text{Sr}$ ) is an unstable nuclide.

The activity of a sample of  $1.0 \times 10^{-9}$  kg of strontium-90 is 5.2 MBq.

- (i) Determine the decay constant  $\lambda$  of strontium-90.

$$\lambda = \dots\dots\dots \text{s}^{-1} \quad [3]$$

- (ii) The activity of the sample after a time of 1.0 half lives is found to be greater than the expected 2.6 MBq.

Suggest a possible reason for this.

.....  
..... [1]

[Total: 6]

52. O/N/21/41/Q12

- (a) Define radioactive *decay constant*.

.....

.....

..... [2]

- (b) A sample of radioactive iodine-131 ( $^{131}_{53}\text{I}$ ) of mass  $5.87 \times 10^{-10} \text{ kg}$  has an activity of  $2.92 \times 10^9 \text{ Bq}$ .

Determine the decay constant of iodine-131.

decay constant = .....  $\text{s}^{-1}$  [3]

- (c) Suggest **two** reasons why a detector placed near to the sample in (b) would record a count rate much less than  $2.92 \times 10^9$  counts per second.

1. ....

.....

2. ....

.....

[2]

[Total: 7]

53. O/N/21/42/Q12

- (a) Radioactive decay is both random and spontaneous.

State what is meant by:

- (i) *random*

.....  
..... [1]

- (ii) *spontaneous*.

.....  
..... [1]

- (b) A sample of radioactive material contains atoms of an unstable nuclide X. The activity of the sample due to the atoms of X is  $A$ . The variation with time  $t$  of  $\ln A$  is shown in Fig. 12.1.

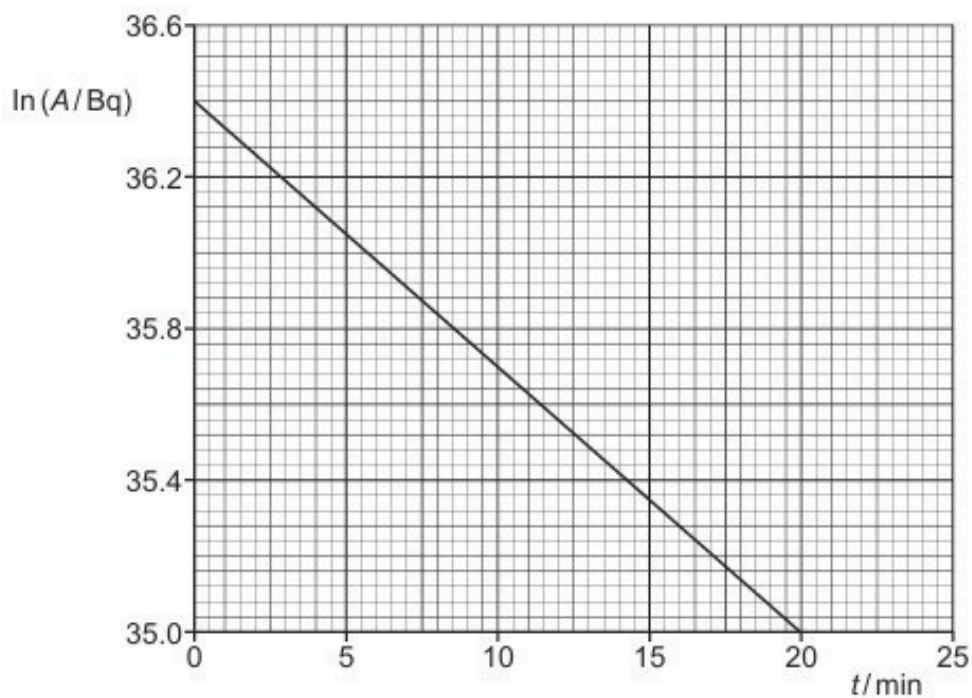


Fig. 12.1

- (i) Use Fig. 12.1 to determine the half-life, in minutes, of nuclide X.

half-life = ..... min [3]

- (ii) At time  $t = 0$ , the mass of the atoms of X in the sample is  $5.66 \times 10^{-7}$  kg.

Determine the nucleon number of X.

nucleon number = ..... [3]

[Total: 8]

54. F/M/22/42/Q9

Polonium-211 ( $^{211}_{84}\text{Po}$ ) decays by alpha emission to form a stable isotope of lead (Pb).

- (a) Complete the equation for this decay.



[2]

- (b) The variation with time  $t$  of the number of unstable nuclei  $N$  in a sample of polonium-211 is shown in Fig. 9.1.

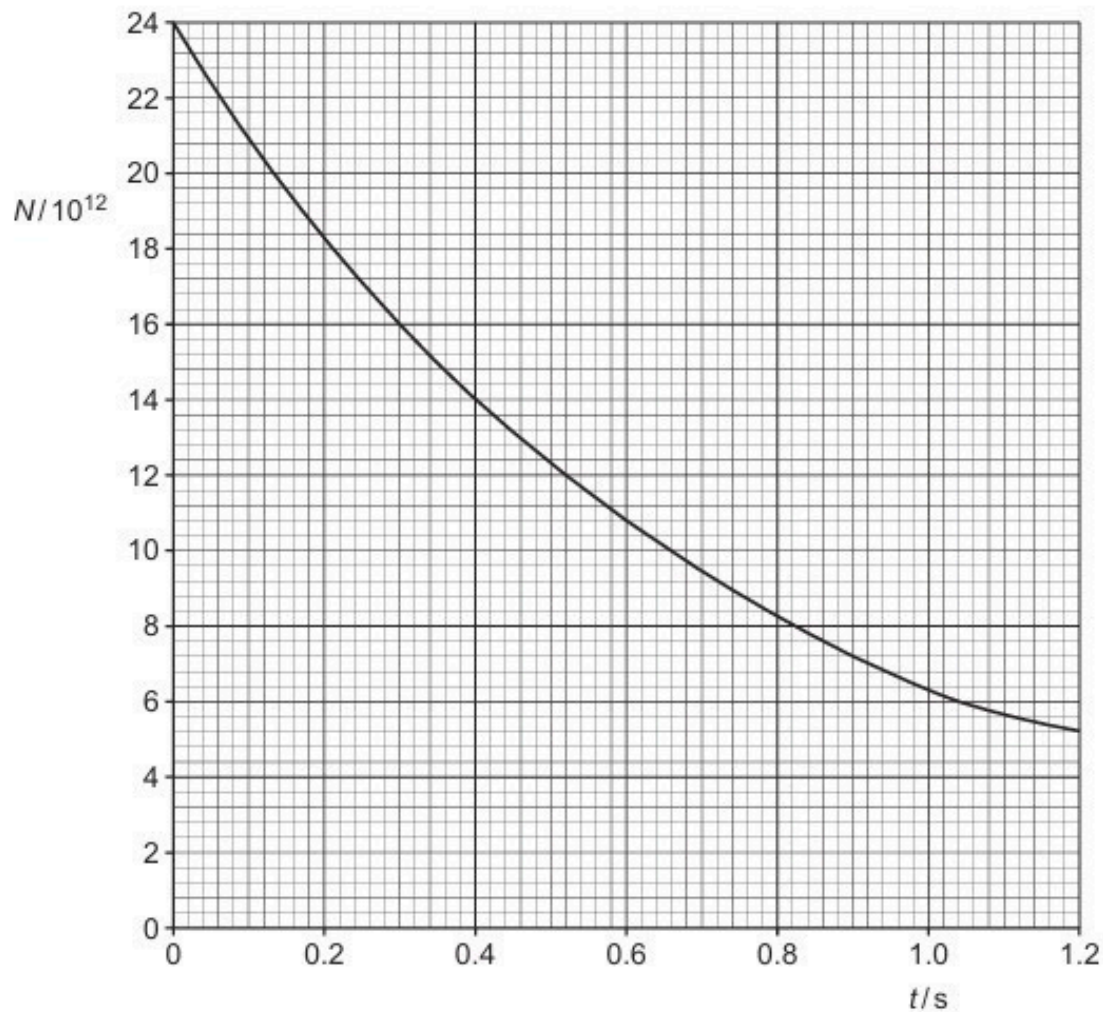


Fig. 9.1

At time  $t = 0$ , the sample contains only polonium-211.

- (i) Use Fig. 9.1 to determine the decay constant  $\lambda$  of polonium-211. Give a unit with your answer.

- (ii) Use your answer in (b)(i) to calculate the activity at time  $t = 0$  of the sample of polonium-211.

activity = ..... Bq [1]

- (iii) On Fig. 9.1, sketch a line to show the variation with  $t$  of the number of lead nuclei in the sample. [2]

- (c) Each decay releases an alpha particle with energy 6900 keV.

- (i) Calculate, in J, the total amount of energy given to alpha particles that are emitted between time  $t = 0.30$  s and time  $t = 0.90$  s.

energy = ..... J [3]

- (ii) Suggest why the total amount of energy released by the decay process between time  $t = 0.30$  s and time  $t = 0.90$  s is greater than your answer in (c)(i).

.....  
.....  
..... [1]

[Total: 11]

55. M/J/22/41/Q8

- (a) (i) State what is meant by nuclear binding energy.

.....

.....

..... [2]

- (ii) On Fig. 8.1, sketch a line to show the variation with nucleon number  $A$  of the binding energy per nucleon  $E$  of a nucleus.

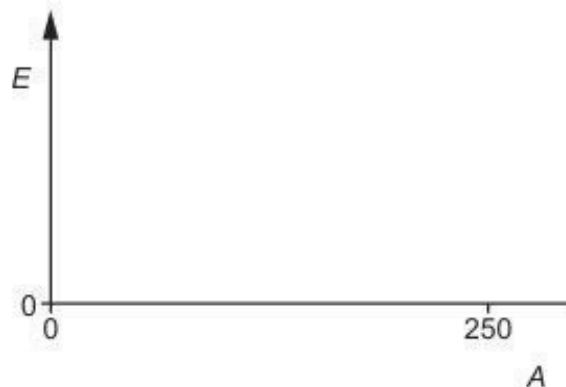


Fig. 8.1

[2]

- (b) In one type of nuclear process, deuterium ( ${}^2_1\text{H}$ ) undergoes the reaction



- (i) State the name of this type of nuclear process.

..... [1]

- (ii) Explain, with reference to your line in (a)(ii), why this reaction results in the release of energy.

.....

.....

..... [2]



(c) Table 8.1 shows the masses of the particles involved in the reaction in (b).

**Table 8.1**

| particle          | mass / u  |
|-------------------|-----------|
| ${}^1_0\text{n}$  | 1.008 665 |
| ${}^2_1\text{H}$  | 2.014 102 |
| ${}^3_2\text{He}$ | 3.016 029 |

Calculate the energy released when 1.00 mol of deuterium undergoes the reaction.

energy = ..... J [5]

[Total: 12]

**56. M/J/22/42/Q10**

(a) State what is meant by radioactive decay.

.....

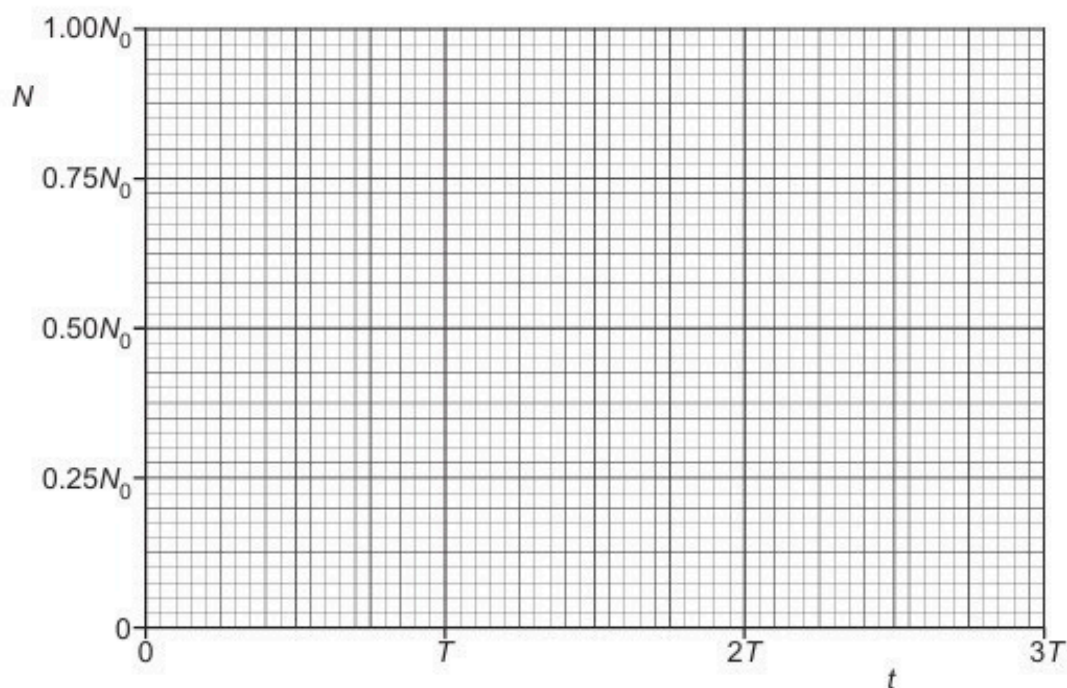
.....

..... [2]

(b) A radioactive sample consists of an isotope X of half-life  $T$  that decays to form a stable product. Only X and the stable product are present in the sample.

At time  $t = 0$ , the sample has an activity of  $A_0$  and contains  $N_0$  nuclei of X.

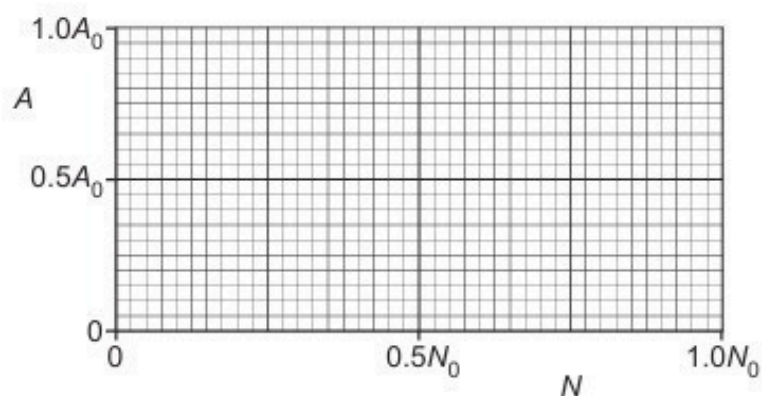
(i) On Fig. 10.1, sketch the variation with  $t$  of the number  $N$  of nuclei of X present in the sample. Your line should extend from time  $t = 0$  to time  $t = 3T$ .



**Fig. 10.1**

[3]

- (ii) On Fig. 10.2, sketch the variation with  $N$  of the activity  $A$  of the sample for values of  $N$  between  $N = 0$  and  $N = N_0$ .



**Fig. 10.2**

[2]

- (c) State the name of the quantity represented by the gradient of your line in:

(i) Fig. 10.1

..... [1]

(ii) Fig. 10.2.

..... [1]

- (d) For the sample in (b), calculate the fraction  $\frac{N}{N_0}$  at time  $t = 1.70T$ .

$$\frac{N}{N_0} = \dots\dots\dots [2]$$

[Total: 11]

**57. O/N/22/41/Q10**

Carbon-15 ( $^{15}_6\text{C}$ ) is an isotope of carbon that undergoes radioactive decay to nitrogen-15 ( $^{15}_7\text{N}$ ), which is a stable isotope of nitrogen.

Radioactive decay is both a random and a spontaneous process.

(a) State what is meant by:

(i) random

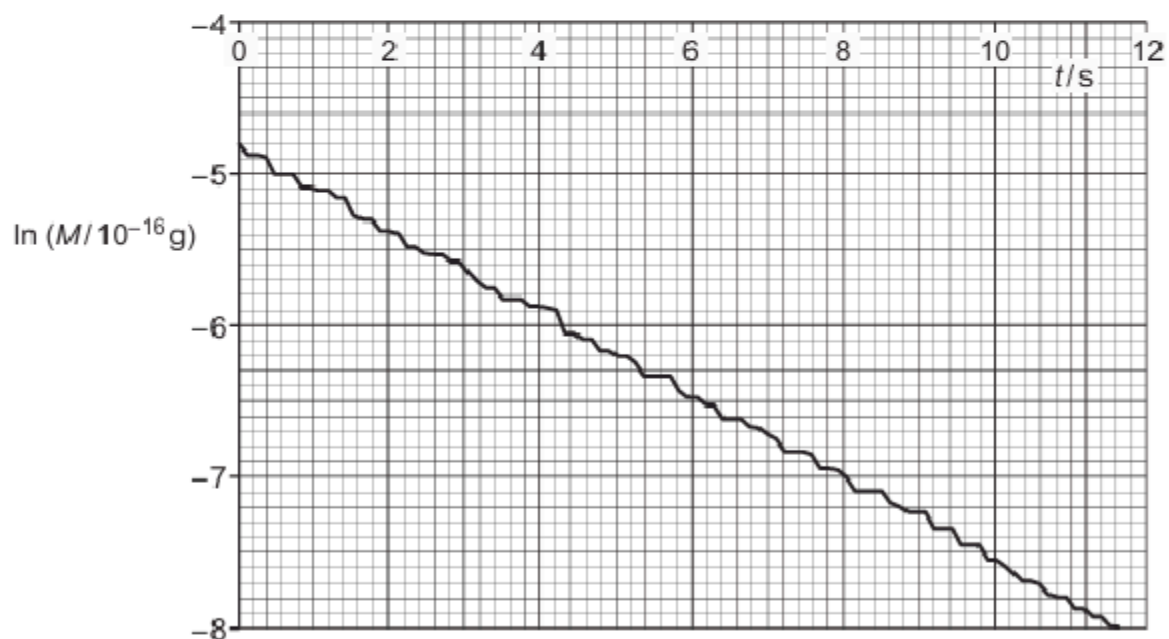
.....  
 ..... [1]

(ii) spontaneous.

.....  
 ..... [1]

(b) A small sample of carbon-15 decays. The mass  $M$  of carbon-15 in the sample decreases with time  $t$ .

Fig. 10.1 shows the variation with  $t$  of the value of  $\ln (M/10^{-16}\text{g})$ .



(i) State how Fig. 10.1 demonstrates that radioactive decay is random.

.....  
 ..... [1]

(ii) On Fig. 10.1, draw the straight line of best fit.

[1]

- (iii) Show that the decay constant  $\lambda$  of carbon-15 is given by the magnitude of the gradient of your line in (b)(ii).

[1]

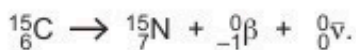
- (iv) Use your line in (b)(ii) to determine  $\lambda$ . Give a unit with your answer.

$\lambda = \dots\dots\dots$  unit  $\dots\dots\dots$  [2]

- (v) Use your answer in (b)(iv) to calculate the half-life of carbon-15.

half-life =  $\dots\dots\dots$  s [1]

- (c) The equation for the decay of carbon-15 can be written as



State and explain how the mass of the products of the decay must compare with the mass of the carbon-15 nucleus.

.....

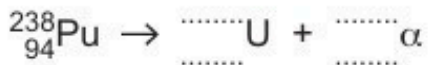
.....

..... [2]

**58. F/M/23/42/Q8**

Plutonium-238 ( $^{238}_{94}\text{Pu}$ ) is unstable and undergoes alpha decay.

- (a) Complete the equation to show the decay of plutonium-238.



[2]

- (b) The power source in a space probe contains 0.874 kg of plutonium-238. Each nucleus of plutonium-238 that decays emits 5.59 MeV of energy. The half-life of plutonium-238 is 87.7 years.

- (i) Calculate the initial number  $N_0$  of nuclei of plutonium-238 in the power source.

$$N_0 = \dots\dots\dots [1]$$

- (ii) Determine the initial activity of the source. Give a unit with your answer.

$$\text{activity} = \dots\dots\dots \text{unit} \dots\dots\dots [2]$$

- (iii) Use your answer in (b)(ii) to determine the initial power output from the source due to the decay of plutonium-238.

$$\text{power output} = \dots\dots\dots \text{W} [2]$$

- (iv) The space probe will continue to function until the power output from the plutonium in the source decreases to 65.3% of its initial value.

Calculate the time, in years, for which the space probe will function.

time = ..... years [2]

- (c) An alternative power source uses energy generated from the radioactive decay of polonium-210. This isotope has a half-life of 0.378 years. The mass of the isotope needed for the same initial power output as in (b) is 3.37 g.

Suggest **one** advantage and **one** disadvantage of using polonium-210 as the source of energy.

advantage .....

.....

disadvantage .....

.....

[2]

[Total: 11]

59. M/J/23/41/Q9

Carbon-11 is radioactive and decays by  $\beta^+$  emission to form boron-11. Carbon-11 has a half-life of 20 minutes. Boron-11 is stable.

- (a) Define half-life.

.....  
 ..... [1]

- (b) A sample contains  $N_0$  nuclei of carbon-11 and no other nuclei at time  $t = 0$ .

On Fig. 9.1, sketch the variation with  $t$  of the number of nuclei of **boron-11** in the sample.

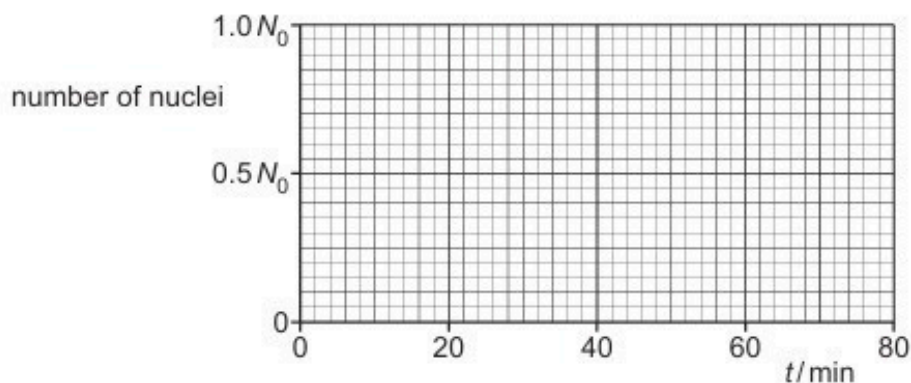


Fig. 9.1

[3]

- (c) (i) Explain, with reference to the random nature of radioactive decay, why the activity of the carbon-11 sample in (b) decreases with time.

.....  
 .....  
 ..... [2]

- (ii) State, with reasons, whether a radiation detector placed near to the sample of carbon-11 indicates a measured count rate from the sample that is less than, the same as or greater than the activity of the sample.

.....  
 .....  
 .....  
 .....  
 ..... [3]

[Total: 9]



**60. M/J/23/42/Q9**

(a) Define mass defect.

.....

.....

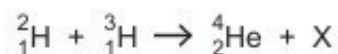
..... [2]

(b) Table 9.1 shows the mass defects of three nuclei.

**Table 9.1**

| nucleus           | mass defect / u |
|-------------------|-----------------|
| ${}^2_1\text{H}$  | 0.002 388       |
| ${}^3_1\text{H}$  | 0.009 105       |
| ${}^4_2\text{He}$ | 0.030 377       |

The nuclear fusion process in a particular star is described by



where X is a particle that has no mass defect.

(i) State the name of particle X.

..... [1]

(ii) Show that the energy released when one nucleus of  ${}^4_2\text{He}$  is formed in this fusion reaction is  $2.8 \times 10^{-12} \text{ J}$ .

- (c) The star in (b) has a radius of  $2.3 \times 10^9 \text{ m}$  and a luminosity of  $1.4 \times 10^{28} \text{ W}$ .  
All the energy released from the formation of  ${}^4_2\text{He}$  is radiated away from the star.  
All the energy that is radiated from the star has been released in the formation of  ${}^4_2\text{He}$ .

Determine:

- (i) the mass of  ${}^4_2\text{He}$  produced per unit time by the fusion process

mass per unit time = .....  $\text{kg s}^{-1}$  [3]

- (ii) the surface temperature of the star.

temperature = ..... K [2]

[Total: 11]

**61. O/N/23/41/Q9**

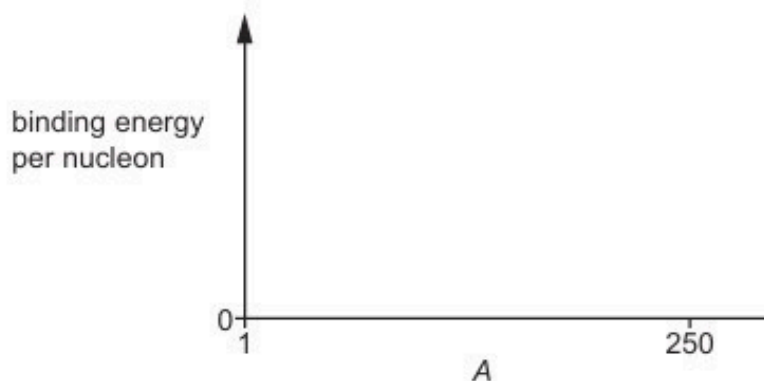
**(a)** State what is meant by nuclear fusion.

.....

.....

..... [2]

**(b)** On Fig. 9.1, sketch the variation of binding energy per nucleon with nucleon number  $A$  for values of  $A$  between 1 and 250.



**Fig. 9.1**

[2]

**(c)** On your line in Fig. 9.1, label:

**(i)** a point X that could represent a nucleus that undergoes alpha-decay [1]

**(ii)** a point Y that could represent a nucleus that undergoes nuclear fusion. [1]

- (d) A nucleus Z undergoes nuclear fission to form strontium-93 ( $^{93}_{38}\text{Sr}$ ) and xenon-139 ( $^{139}_{54}\text{Xe}$ ) according to

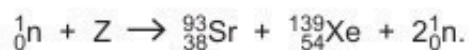


Table 9.1 shows the binding energies of the strontium-93 and xenon-139 nuclei.

**Table 9.1**

| nucleus                | binding energy/J       |
|------------------------|------------------------|
| $^{93}_{38}\text{Sr}$  | $1.25 \times 10^{-10}$ |
| $^{139}_{54}\text{Xe}$ | $1.81 \times 10^{-10}$ |

The fission of 1.00 mol of Z releases  $1.77 \times 10^{13}$  J of energy.

Determine the binding energy per nucleon, in MeV, of Z.

binding energy per nucleon = ..... MeV [4]

[Total: 10]